

Unit 1 - Basics of Geometry and Constructions

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The following B.E.S.T Standards will be covered in this unit:

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

- *Clarification 1: Postulates, relationships, and theorems including vertical angles are congruent; when a transversal crosses parallel lines, the consecutive angles are supplementary and alternate (interior and exterior) angles and corresponding angles are congruent; points on a perpendicular bisector of a line segment are exactly those equidistant from the segment's endpoints.*
- *Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flowchart proofs or informal proofs.*
- *Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.*

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

- *Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.*
- *Clarification 2: Instruction includes recognizing that these transformations preserve angle measure.*

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

- *Clarification 1: Transformations include translations, dilations, rotations and reflections described using words or using coordinates.*
- *Clarification 2: Within the Geometry course, figures are limited to triangles and quadrilaterals and rotations are limited to 90° , 180° , and 270° counterclockwise or clockwise about the center of rotation.*
- *Clarification 3: Instruction includes the understanding that when a figure is mapped onto itself using a reflection, it occurs over a line of symmetry.*

MA.912.GR.2.4 (Honors only) Determine symmetries of reflection, symmetries of rotation and symmetries of translation of a geometric figure.

- *Clarification 1: Instruction includes determining the order of each symmetry.*
- *Clarification 2: Instruction includes the connection between tessellations of the plane and symmetries of translations.*

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.

- *Clarification 1: Instruction includes showing that the corresponding sides and the corresponding angles are congruent.*

MA.912.GR.2.8 Apply an appropriate transformation to map one figure onto another to justify that the two figures are similar.

- *Clarification 1: Instruction includes showing that the corresponding sides are proportional, and the corresponding angles are congruent.*

MA.912.GR.5.1 Construct a copy of a segment or an angle.

- *Clarification 1: Instruction includes using compass and straightedge, string, reflective devices, paper folding, or dynamic geometric software.*

MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.

- *Clarification 1: Instruction includes using compass and straightedge, string, reflective devices, paper folding, or dynamic geometric software.*

Unit 1, Lesson 1: Defined and Undefined Terms – Part 1



Benchmarks

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.



Learning Targets

- ☐ I can use rigid motions to define parallel lines and perpendicular lines.
- ☐ I can experiment with rigid motions on a plane.
- ☐ I can use rigid motions to define a right angle.



1.1.1 Warm-Up: A Billion!

One billion is a big number. Starting the school year, it may feel like there are a billion seconds left in the year. When you started school there were 1.555×10^7 , or 15,550,000 seconds left in the school year.

A billion seconds is approximately 32 years!

1000000000	=	31.7097919838
Second		Calendar year

So, although it may feel like a billion, it really isn't.

1. What is the location of 1 billion on the line segment?



2. Describe your strategy for estimating the location of 1 billion on the line segment.



1.1.2 Guided Instruction: Definitions

1. In the Warm-Up, the diagram is a line segment.
 - a. Discuss with your partner why it wasn't called a line.
 - b. Write your best definition of a line segment.
 - c. Draw a line. Label the line using the letter h .
 - d. Write your best definition of a line.

- e. Ask another partner pair the definitions they wrote. Record their definition and write your summary of their reason for the definition. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Definition	My Summary of Their Reason	Initials
A line segment is...		
A line is...		



Facts About Lines

- A line has infinite length, with no height and no width.
- A line is assumed to be straight.
- A line is considered one-dimensional.
- A line is considered an undefined term.
- A line goes on forever in both directions.

A line has infinite length, thus, any line drawn should have arrowheads on both ends.

A line can be named by a single lowercase script letter, or by any two points which lie on the line.

2. Consider the two lines shown.



- Use the script lowercase letter m to name the purple line on the left.
- Use the letters H and R to name the orange line on the right.



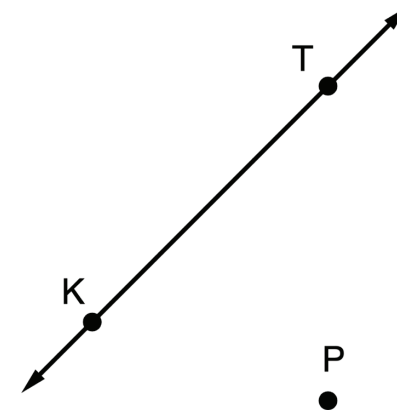
Facts About Planes

- A plane has infinite length and width, with no height.
- A plane is considered two-dimensional.
- A plane is considered an undefined term.
- A plane goes on forever.

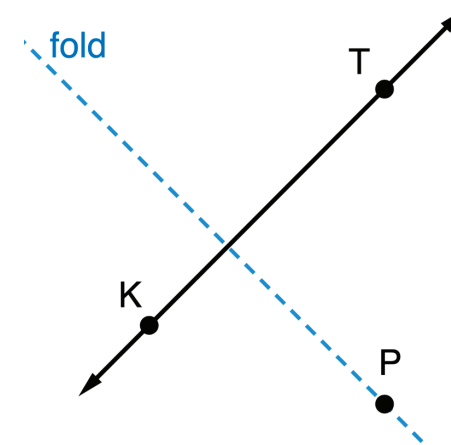
A plane is named by a single uppercase letter written in script or by three points that do not lie on the same line.



3. Consider the line and point shown.

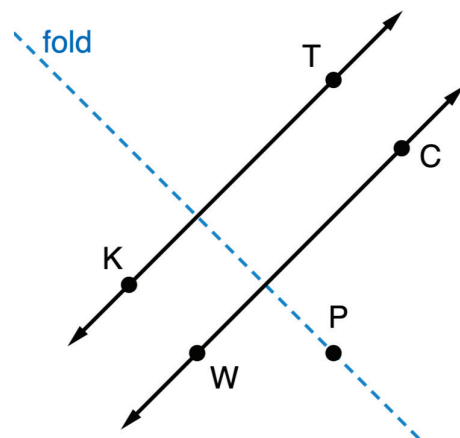


- Complete the following:
 - On a piece of patty paper, copy line KT and point P .
 - Fold the patty paper so that the fold goes through point P and each part of line KT coincides with itself. Unfold the patty paper and draw a dashed line along the fold through the point, as shown.

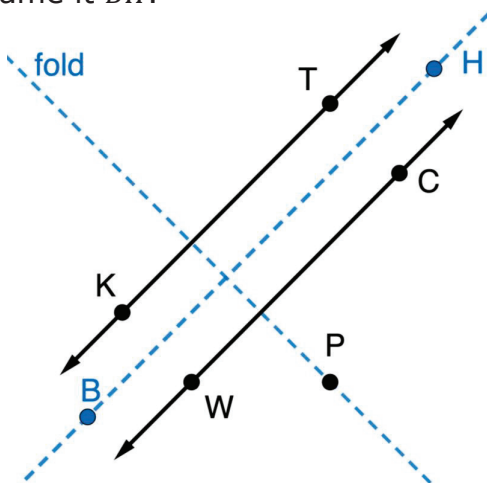


- Fold the dashed line on itself so that point P coincides with $\overleftrightarrow{KT}$.

- The resulting fold is a line that is parallel to the $\overleftrightarrow{KT}$. Use a writing utensil and straightedge to draw the parallel line. Label the line using letters W and C as shown.



- Show the lines are parallel by folding the patty paper so that $\overleftrightarrow{KT}$ and $\overleftrightarrow{WC}$ coincide. Unfold the paper and draw a dashed line along the fold and name it $\overleftrightarrow{BH}$.



b. Complete the statement.

$\overleftrightarrow{WC}$ is a
☐ rotation
☐ reflection
 of $\overleftrightarrow{KT}$.

- Work with your partner to describe how the lines you drew on patty paper, $\overleftrightarrow{KT}$ and $\overleftrightarrow{WC}$, lie on the same plane.



Parallel lines do not intersect and lie in the same plane.

The table shows multiple examples of parallel lines.

Description	Example of Parallel Lines
Parallel lines $\overleftrightarrow{TW}$ and $\overleftrightarrow{KD}$ are shown in plane C .	
Parallel lines $\overleftrightarrow{TW}$ and $\overleftrightarrow{KD}$ are shown in planes B and N .	
Parallel lines $\overleftrightarrow{TW}$ and $\overleftrightarrow{KD}$ are shown in planes B, C , and N .	



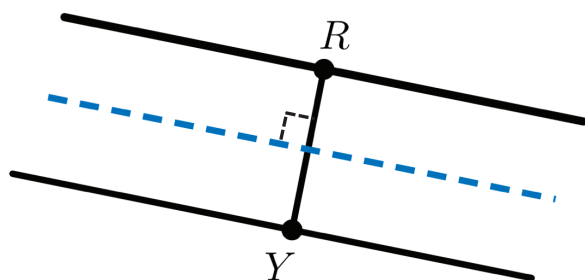
1.1.3 Collaboration: Definitions

1. Continue using the patty paper from the previous component.
 - a. Label a point on $\overleftrightarrow{KT}$ as R . Then, fold the patty paper along $\overleftrightarrow{BH}$ and copy point R to $\overleftrightarrow{WC}$ naming it Y . Unfold the paper and draw a line segment that connects points R and Y .
 - b. What type of angle does $\overline{RY}$ form with the dashed line?



A **right angle** is an angle measuring exactly 90° .

- c. Use a ruler to measure the distance between point R and the dashed line, and between point Y and the dashed line. Write the measurements, with units, on the diagram.



- d. Discuss with your partner: when you reflect an object across a line, will corresponding points be the same distance from the line of reflection? Summarize your discussion.

2. Complete the statements.

The patty paper is similar to a ☐ plane.
☐ reflection.

By reflecting a line across a line of ☐ reflection,
☐ rotation,
a parallel line was created.

The parallel line ☐ is
☐ is not the same distance

from the line of ☐ reflection.
☐ rotation.

3. How is $\overline{RY}$ related to the dashed line?



Perpendicular lines are lines that intersect at right angles.

The table shows multiple examples of perpendicular lines.

Description	Example of Perpendicular Lines
Perpendicular lines $\overleftrightarrow{KT}$ and $\overleftrightarrow{RB}$ are shown in plane X .	
Perpendicular lines $\overleftrightarrow{KT}$ and $\overleftrightarrow{RB}$ are shown in planes M and W .	
Perpendicular lines $\overleftrightarrow{KT}$ and $\overleftrightarrow{RB}$ are shown in planes M , W , and X .	

4. If two lines are perpendicular, then any two adjacent angles formed are _____.



Skew lines are lines that do not intersect and are not parallel.

The table shows an example of skew lines.

Description	Example of Skew Lines
Skew lines $\overleftrightarrow{HB}$ and $\overleftrightarrow{KN}$ are shown in planes L and S .	



1.1.4 Wrap-Up: Reflection

1. Rate yourself based on what you understood in today's lesson.

Skill	1 – I am just starting. 2 – I have some skills. 3 – I am almost there. 4 – I have it!
I can define parallel lines.	
I can define perpendicular lines.	
I can define a right angle.	

2. Rate yourself based on how you feel about the following statements.

Statement	1 – I struggle with this. 2 – I am ready to grow in this area. 3 – I am almost there. 4 – That's me!
I am willing to try new things.	
I persevere when something is challenging.	



Unit 1, Lesson 2: Defined and Undefined Terms – Part 2



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.



Learning Targets

- ☐ I can understand the three undefined terms of geometry.
- ☐ I can use rigid motions to define a straight angle.
- ☐ I can experiment with rigid motions on a plane.



1.2.1 Warm-Up: Would You Rather?

1. Would you rather have pennies to match your weight or quarters to match your height?
2. Explain your reasoning.



1.2.2 Exploration: Definitions

1. Use two sheets of patty paper to complete the following.
 - Draw $\overline{HN}$ on one piece of patty paper.
 - Draw $\overline{BL}$ on the other piece of patty paper.
 - Lay the piece of patty paper with $\overline{HN}$ drawn on it on top of the piece with $\overline{BL}$ drawn on it. Place the papers so that the points H and B are on top of each other and the line segments coincide.
 - Place your writing utensil on endpoint H .
 - Rotate the top sheet of patty paper until the two line segments make an obtuse angle.
2. Draw the obtuse angle you created below.
3. When you rotated the top sheet, did the line segment change in size?
4. When you rotated the top sheet, did the line segment change in position?

5. Which part(s) of the obtuse angle has two points?
6. Rotate the top sheet of patty paper until the two line segments make a straight line.
7. Name the resulting line segment using the endpoints.
8. Work with your partner to estimate the number of degrees you rotated the top sheet of patty paper from its original position on top of line segment BL .



A **straight angle** is an angle measuring exactly 180° .

9. Based on the rotation and the definition of a straight angle, what “fact” about a line can be added to the “Facts about Lines” in the previous lesson?



1.2.3 Bring It Together: Undefined Terms



A **definition** is an exact statement, written formally, of the meaning or description of a word.

There are three terms in geometry that are considered “undefined terms.”

A term is considered undefined if, when explaining the term, the explanation relies on the term itself or a close synonym. This is sometimes called a circular definition.

1. In the previous lesson you wrote a definition for the term “line.” Review your definition and determine if your definition is considered circular.

Although we have “facts about lines,” “line” is an undefined term in geometry. The other two undefined terms in geometry are “point” and “plane.” Look back through the previous lesson to notice how often these terms were used without providing a definition.

2. Work with your partner to label each description as a line, plane, or point.

Description	Undefined Term
has two dimensions	☐ Line ☐ Point ☐ Plane
has no length, no width, and no height	☐ Line ☐ Point ☐ Plane
is often drawn as a four-sided figure	☐ Line ☐ Point ☐ Plane
has infinite length, no width, no height	☐ Line ☐ Point ☐ Plane
named by an ordered pair in the coordinate plane	☐ Line ☐ Point ☐ Plane
indicates a location or position	☐ Line ☐ Point ☐ Plane
has one dimension	☐ Line ☐ Point ☐ Plane
has infinite length, infinite width, and zero height	☐ Line ☐ Point ☐ Plane
has no dimension or actual size	☐ Line ☐ Point ☐ Plane



1.2.4 Wrap-Up: Reflection

1. Rate yourself based on what you understood in today's lesson.

Skill	1 – I am just starting. 2 – I have some skills. 3 – I am almost there. 4 – I have it!
I can understand the three undefined terms of geometry.	
I can define a straight angle.	

2. Rate yourself based on how you feel about the following statements.

Statement	1 – I struggle with this. 2 – I am ready to grow in this area. 3 – I am almost there. 4 – That's me!
I am a student learner.	
I am a mathematician.	

Unit 1, Lesson 3: Isometry



Benchmark

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.



Learning Targets

- ☐ I can define rigid motions.
- ☐ I can show rigid motions by mapping using arrows on the plane.
- ☐ I can explain why isometry preserves distance.



1.3.1 Warm-Up: Animation Flipbook

An animation flipbook is a small book of drawings with a series of pictures that appear to move when the pages are flipped in rapid succession.



The video shows a series of pictures of two clouds. A few screenshots are shown.



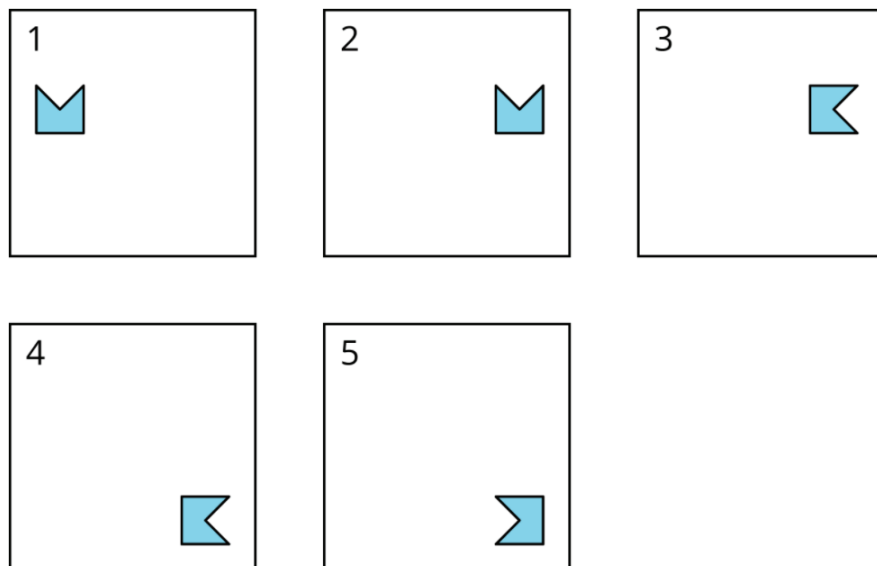
1. What do you notice about the two clouds?
2. At one point the two clouds seemed to move up and down. How do you think the artist was able to do that?



1.3.2 Exploration: Isometry

Work with your partner to complete the following.

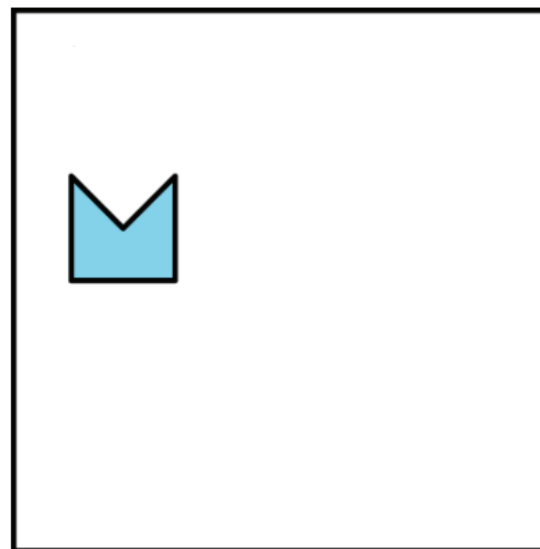
- Five frames that feature a polygon in different positions are shown.



Describe how the polygon moves from one frame to the next.

Movement from...	Description
Frame 1 to Frame 2	
Frame 2 to Frame 3	
Frame 3 to Frame 4	
Frame 4 to Frame 5	

- The diagram shows the position of the polygon from Frame 1.



- Label two pieces of patty paper as **A** and **B**. Then, complete the steps for each piece of patty paper.

Patty paper A

- Copy the polygon.
- Move patty paper **A** to display the polygon as it is shown in Frame 2.

Patty paper B

- Copy the polygon and the square that surrounds it.
- Fold the patty paper vertically so that the top corners of the square match up.
- With the patty paper folded, draw the copy of the polygon onto the other half of the patty paper.

b. Complete the statements.

To draw the polygon as it is shown in Frame 2,

patty paper **A** was moved

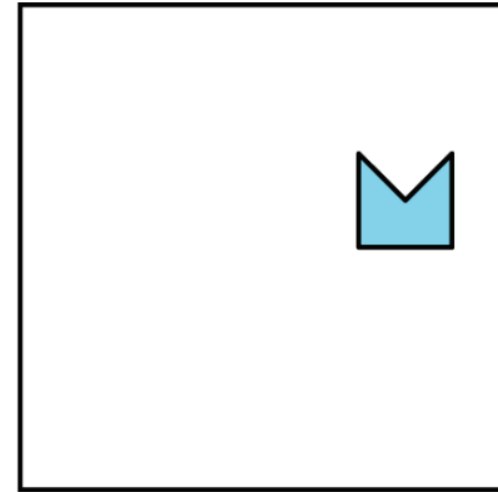
- ☐ up.
- ☐ left.
- ☐ right.
- ☐ down.

When patty paper **B** was folded, the polygon could be traced so that the new polygon is as shown in Frame 2.

The original polygon was ☐ flipped across ☐ rotated around the fold of the paper.

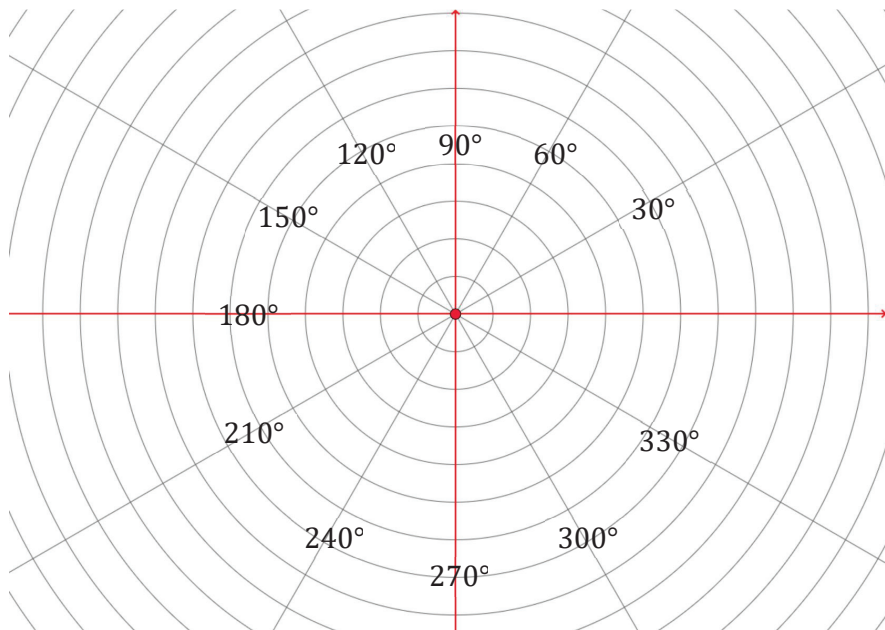
The fold of the paper was the ☐ center of rotation. ☐ line of reflection.

3. The diagram shows where the polygon is in Frame 2.



a. Complete the steps to find the point of rotation.

- Label a piece of patty paper as **C**.
- Copy the polygon onto patty paper **C**.
- Use the point of your writing utensil to find the point of rotation so the polygon can be rotated to look like Frame 3.
- Place a dot on the diagram above to show the point of rotation.
- Place patty paper **C** on the circular graph paper so the point of rotation is where the two red axes meet, and the polygon is oriented as in Frame 2.
- Rotate patty paper **C** so that it looks like Frame 3.



b. Complete the statements.

To move the polygon so it looks like Frame 3, rotate the diagram on patty paper **C** _____ degrees clockwise ↻ or _____ degrees counterclockwise ↺.

The center of rotation is

- the top left vertex.
- the top right vertex.
- the bottom left vertex.
- the bottom right vertex.
- the center of the polygon.

4. Use patty paper, a ruler, or a protractor as necessary to complete the following.

a. Complete the table by determining if the polygon changed in size or shape as it moved.

Movement from...	Did the size change?	Did the shape change?
Frame 1 to Frame 2		
Frame 2 to Frame 3		
Frame 3 to Frame 4		
Frame 4 to Frame 5		

b. What tools or information were used when determining if the size changed?

c. What tools or information were used when determining if the shape changed?

d. Discuss with your partner if the measurement of angles in the polygon changed from Frame 1 to Frame 5.



1.3.3 Bring It Together: Definitions

The movements of the polygon in the five frames are known as reflection, rotation, and translation. During the lesson a reflection was referred to as a flip, which is an informal definition. Another informal definition used was “is moved,” which is formally known as a translation.



A **reflection** is a transformation that produces the mirror image of a geometric figure across a line of reflection.



A **rotation** is a transformation of a figure by turning it about a center point or axis. The amount of rotation can be expressed in the number of degrees. The direction of the rotation for two-dimensional figures can be expressed as clockwise or counterclockwise.



A **translation** is a transformation in which every point in a figure is moved in the same direction and by the same distance.

Isometry is a transformation that preserves distance and angle measure.

1. Discuss with your partner the definition of isometry. Determine which transformations — reflection, rotation, or translation — preserve distance and angle measure. Summarize your discussion.



A **rigid transformation** is a transformation of points in space consisting of a sequence of one or more translations, reflections, or rotations. Rigid transformations preserve distances and angle measures (congruency).



A rigid transformation is also called a **rigid motion**. Rigid means that something is not able to be changed. Isometry is composed of rigid motions.

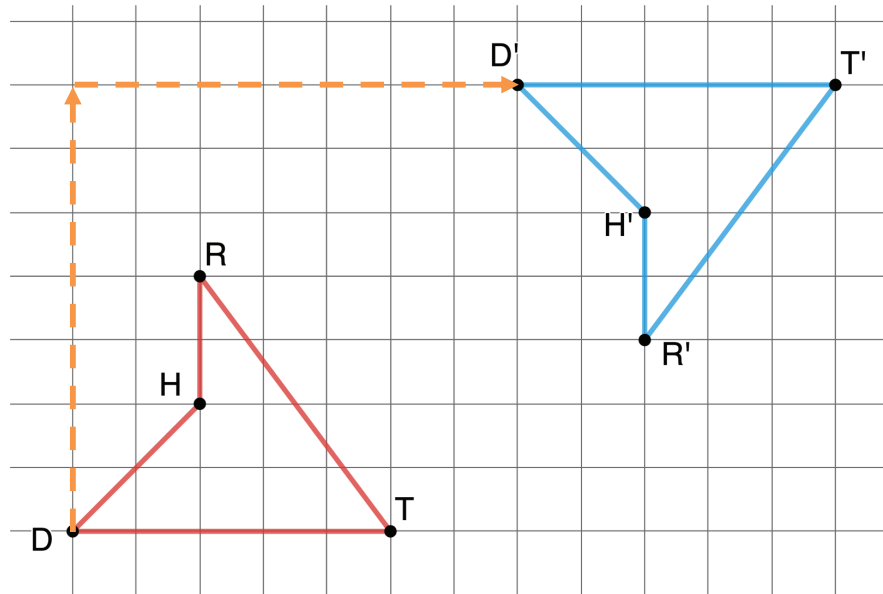


Figures having exactly the same shape and size are **congruent**. Equivalently, two figures are congruent if one can be mapped to the other using a rigid transformation.



1.3.4 Collaboration: Isometry

1. A diagram of $RTDH$ and $R'T'D'H'$ is shown. Arrows are drawn from vertex D to vertex D' .



$R'T'D'H'$ is read as "r prime, t prime, d prime, h prime". This notation is often used when a figure has been transformed.

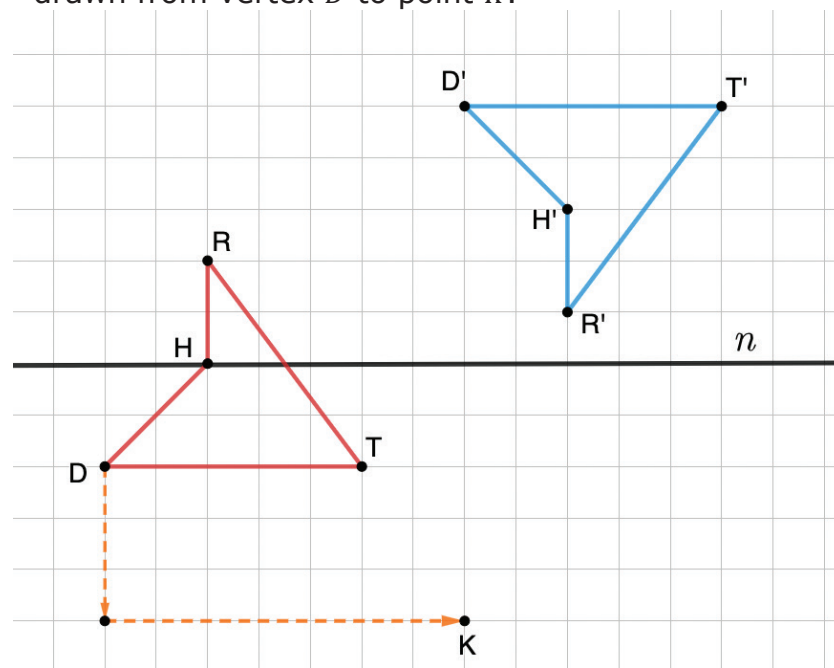
The arrows show a possible path of movement. For the movement to be correct, the same path should exist for all points on a diagram.

Discuss with your partner if all of the points on $RTDH$ can be moved using the same path as point D .



A **mapping** is a standardized form of showing a transformation.

2. A diagram of $RTDH$ and $R'T'D'H'$ is shown. Arrows are drawn from vertex D to point K .



- Draw a line segment from point K to line n . What is the length of the line segment?
- Draw a line segment from point D' to line n . What is the length of the line segment?
- Describe the relationship each line segment has to line n .



If corresponding points have the same distance from a line, then the line is the “**line of reflection.**”

- d. For each of the other vertices on $RTDH$, draw a new point using the same movements done to map point D to point K . Then show the reflection to $R'T'D'H'$ using line segments.
- e. Discuss with your partner if the mapping of **all** of the points on $RTDH$ can be done using the same path and reflection, or if the mapping only works for the vertices. Summarize your discussion.



1.3.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the exploration that I knew were ...	Something I wondered during the exploration was ...	Something I learned during the exploration was ...

Unit 1, Lesson 4: Dilations and Partitions



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.GR.2.2 Identify transformations that do or do not preserve distance.



Learning Targets

- ☐ I can dilate a line segment.
- ☐ I can define midpoint using dilation and understand partitions.
- ☐ I can relate scale factor to dilation.



1.4.1 Warm-Up: Floyd Norman

Floyd Norman was named a Disney Legend in 2007. In 1957, Norman was employed as an inbetweener on *Sleeping Beauty*. An inbetweener is an animator who draws the intermediate frames between two key frames. The inbetweener's drawings create the illusion of motion.

When asked, "What is the key to growing as an artist?" Norman said, "As a student, I think the most important thing is having a willingness to learn."

1. On a scale of 1 to 10, how willing are you to learn?

When asked about his career Norman stated, "Making mistakes is how we grow, how we learn." He said he had no confidence, but at least he made the first step.

2. Describe a time in your life when you had no confidence, but you made the first step and learned from making mistakes.

Norman described a time when Walt Disney asked him to write. Norman made these two statements as he reflected on doing something he did not think he could do.

- "The only obstacle you face is yourself."
- "If you are willing to do the work, make that commitment, then I don't believe there is an obstacle."

3. What subject(s) are obstacles for you?



1.4.2 Exploration: Dilating a Line Segment

1. The diagram shows points W , W' , and W'' .



- What do you notice about the points?
- Estimate the number of times the distance between W and W' will fit between W' and W'' .

- c. Two students described the relationship $\overline{WW''}$ has with $\overline{WW'}$.

Jaquaze	Andrés
$\overline{WW''}$ is three times as long as $\overline{WW'}$.	$\overline{WW''}$ is four times as long as $\overline{WW'}$.

Discuss with your partner which student is correct and the mistake the other student made. Write a brief note to the student who made the mistake.

Means

I

Start

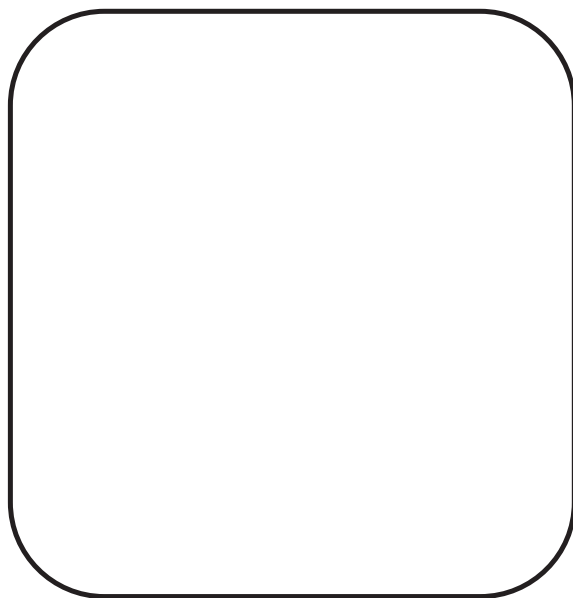
To

Acquire

Knowledge

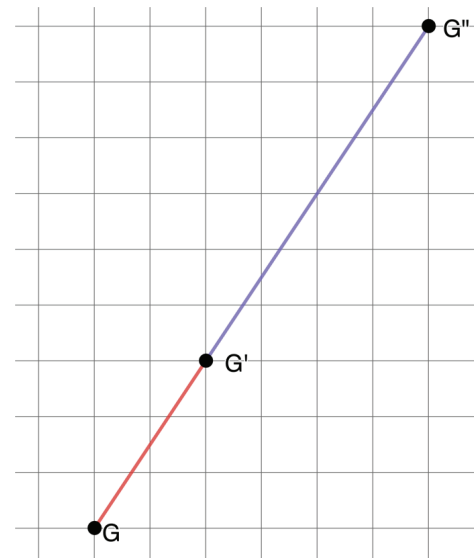
Experience

Skills



2. The diagram shows points G , G' , and G'' .

- a. Use patty paper to determine the number of copies of $\overline{GG'}$ that will fit on $\overline{GG''}$.



- b. Discuss with your partner how to use the grid to determine the number of copies of $\overline{GG'}$ that will fit on $\overline{GG''}$. Summarize your discussion.

- c. Ask a classmate how they would use the grid to determine the number of copies of $\overline{GG'}$ that will fit on $\overline{GG''}$. Record their description and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your description and summary are correct.

Description	My Summary of Their Reason	Initials



A **dilation** is a proportional increase or decrease in size in all directions.

- d. Use the definition of dilation to describe why $\overline{GG''}$ is a dilation of $\overline{GG'}$.

- e. Fill in the blanks to complete the statements.

G' divides $\overline{GG''}$ into a ratio.

The ratio of $\frac{GG'}{GG''}$ is $\frac{\boxed{}}{\boxed{}}$.



Partition means to separate or to divide. A line segment can be partitioned into smaller segments. A ratio can be used to describe the partition.

- f. Use the Pythagorean theorem to complete the table.

Segment	Length
GG'	
$G'G''$	
GG''	

g. Point G' is $\frac{2}{3}$ of the distance from G to G'' .

h. What value can be multiplied to the length of GG' to determine the length of GG'' ?

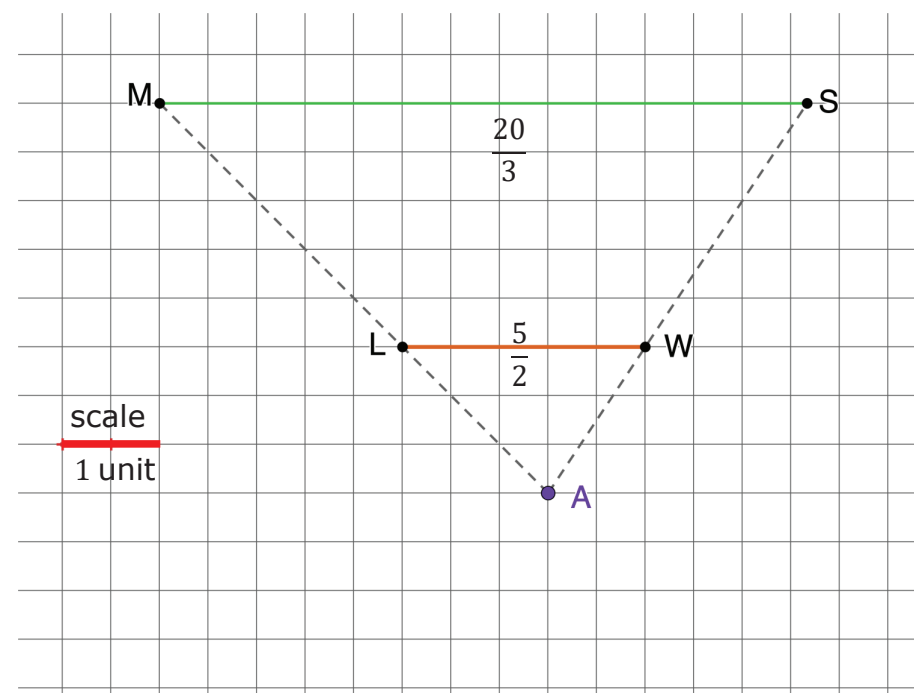


The **scale factor** is the constant that is multiplied by the length of each side of a figure to produce an image that is the same shape as the original figure.



1.4.3 Guided Practice: Scale Factor

$\overline{LW}$ is dilated to create $\overline{MS}$. The center of dilation is point A .



1. $\overline{LW}$ has a length of $\frac{5}{2}$ units and $\overline{MS}$ has a length of $\frac{20}{3}$ units. What is the scale factor that was used to create $\overline{MS}$?

2. Use the Pythagorean theorem to complete the table.
Round values to the nearest tenth of a unit.

Segment	Length	Segment	Length
$\overline{AL}$		$\overline{AW}$	
$\overline{LM}$		$\overline{WS}$	3.0
$\overline{AM}$		$\overline{AS}$	

3. Point L is $\frac{\boxed{}}{\boxed{}}$ of the distance from A to M .

4. What value can be multiplied to the length of $\overline{AL}$ to determine the length of $\overline{AM}$?

5. Point W is $\frac{\boxed{}}{\boxed{}}$ of the distance from A to S .

6. What value can be multiplied to the length of $\overline{AW}$ to determine the length of $\overline{AS}$?

7. What do you notice about each of the following ratios?

- $LW:MS$

- $AL:AM$

- $AW:AS$

8. Work with your partner to explain how to find the scale factor of a dilation.





1.4.4 Bring It Together: Determining Scale Factor

As the class shares strategies for finding scale factor, choose a strategy other than yours to describe.

Strategy	I Liked the Strategy Because...

1. $\overline{XR}$ and $\overline{RK}$ are shown.



a. What is the scale factor that can be used to dilate $\overline{XR}$ to create $\overline{XK}$?

b. What is the scale factor that can be used to dilate $\overline{XK}$ to create $\overline{XR}$?

c. Describe point R in relation to $\overline{XK}$.

The **midpoint** of a line segment is the point that divides a line segment into two equal segments.



1.4.5 Wrap-Up: Cool Down

1. Isometry is a transformation that preserves distance and angle measure. Write a short note that describes whether a dilation preserves distance or not.



Unit 1, Lesson 5: Transformations on the Plane – Part 1



Benchmark

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.



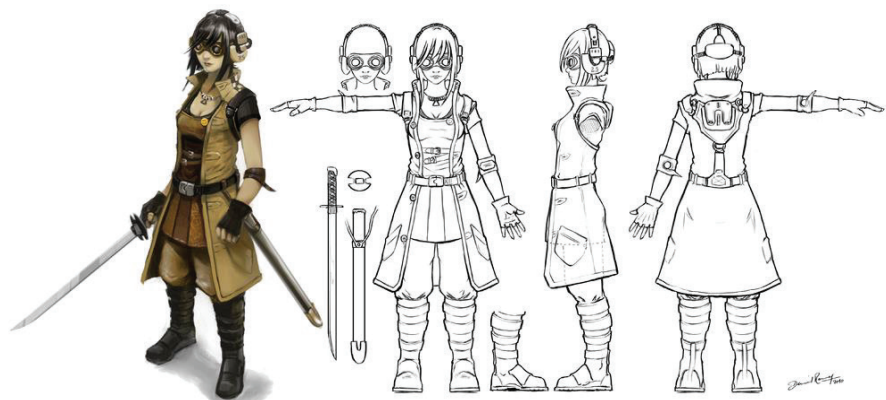
Learning Targets

- ☐ I can identify the angle of rotation of a transformation of a figure on a plane.
- ☐ I can identify the center of rotation of a transformation of a figure on a plane.
- ☐ I can identify the direction of a rotation of a transformation of a figure on a plane.
- ☐ I can distinguish a rotation from other transformations.



1.5.1 Warm-Up: Rotations in Drawings

The picture shows the front, side, and back view of an animated character.



1. What do you notice about all three views — front, side, and back, of the animated character?

When animators and cartoonists sketch the movement of a character, they use a sheet like the one shown.

Front	Side	Back

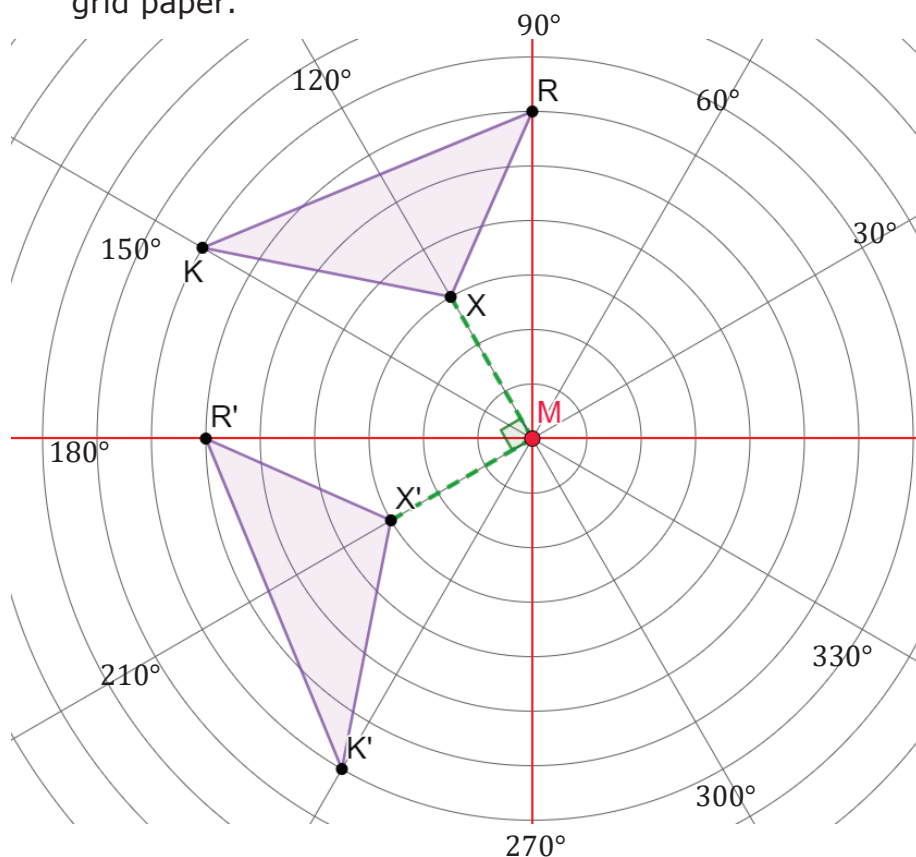
2. Use one of the pictures below, or your own idea, to sketch the three views. Use the dotted lines to help you align key features of the picture.





1.5.2 Guided Instruction: Transformations on the Plane

1. The diagram shows $\triangle RKX$ and $\triangle R'K'X'$ on circular grid paper.



Follow the steps below.

- Use a piece of patty paper to copy $\triangle RKX$ and point M .
- Place the patty paper so that it aligns with $\triangle RKX$ and point M .
- Place the point of your writing utensil on point M and rotate $\triangle RKX$ so that it lays on top of $\triangle R'K'X'$.

a. Draw $\angle RMR'$.

b. Draw $\angle KMK'$.

c. Complete the table.

Angle	Measurement (degrees)
$\angle XMX'$	
$\angle RMR'$	
$\angle KMK'$	

- d. Discuss with your partner how the angle measures on the circular grid can be used to verify the rotation and how to determine the direction of rotation. Summarize your discussion.

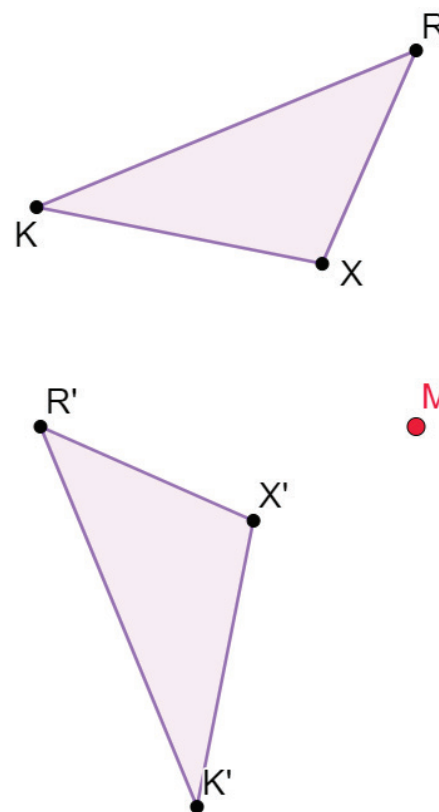
- e. Identify the transformation of $\triangle RKX$. Describe the transformation using direction, degrees, or scale factor.

Type of Transformation	Center, Line, or Movement	Description
○ Dilation ○ Rotation ○ Reflection ○ Translation	○ Center of dilation is _____. ○ Center of rotation is _____. ○ Line of reflection is _____. ○ Down ○ Left ○ Right ○ Up	

- f. Discuss with your partner whether or not the transformation could be described as "a rotation of 270° clockwise centered at point M ." Explain why or why not.

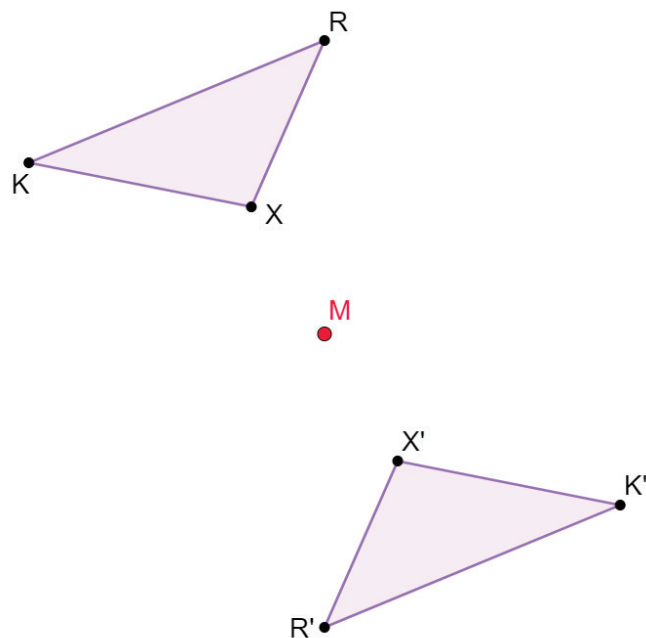
2. The diagram shows $\triangle RKX$ and $\triangle R'K'X'$ without the circular grid paper.

- a. Draw $\angle XMX'$, $\angle RMR'$, and $\angle KMK'$.



- b. Discuss with your partner how the degree of rotation can be determined using the angles that connect the vertices and the center of rotation.

3. The diagram shows $\triangle RKX$ and $\triangle R'K'X'$ using a different transformation.



- a. Identify the transformation of $\triangle RKX$. Describe the transformation using direction, degrees, or scale factor.

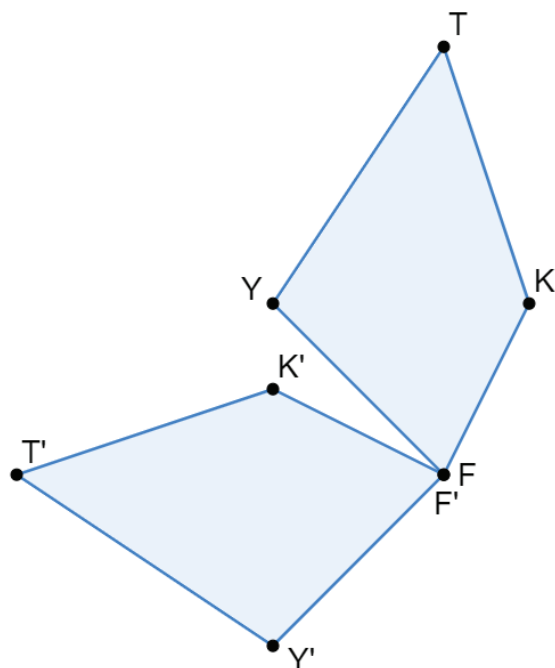
Type of Transformation	Center, Line, or Movement	Description
○ Dilation ○ Rotation ○ Reflection ○ Translation	○ Center of dilation is _____. ○ Center of rotation is _____. ○ Line of reflection is _____. ○ Down ○ Left ○ Right ○ Up	

- b. Two students' descriptions of the transformation of $\triangle RKX$ are shown.

Michela	Marc
a reflection across line M	a translation down

- Use a piece of patty paper to copy $\triangle RKX$.
 - Perform each student's transformation to prove that each transformation does not work.
- c. Discuss with your partner why both students are wrong. Write your reasoning for why the transformations do not work.

d. The diagram shows $TKFY$ and $T'K'F'Y'$.



Identify the transformation of $TKFY$. Describe the transformation using direction, degrees, or scale factor.

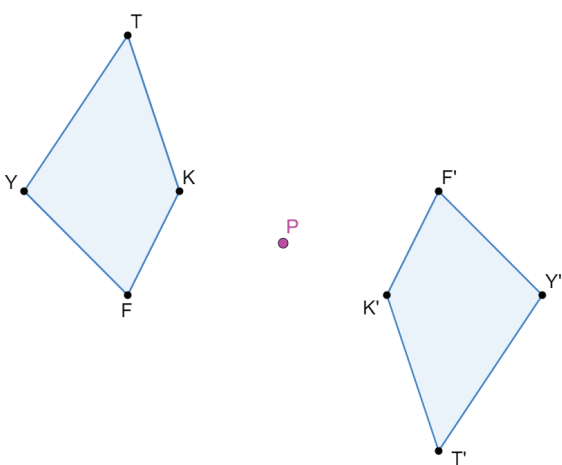
Type of Transformation	Center, Line, or Movement	Description
○ Dilation ○ Rotation ○ Reflection ○ Translation	○ Center of dilation is _____. ○ Center of rotation is _____. ○ Line of reflection is _____. ○ Down ○ Left ○ Right ○ Up	



1.5.3 Guided Practice: Rotations

The table shows three different rotations of $TKFY$. Determine the direction of rotation, center of rotation, and the degree of rotation for each.

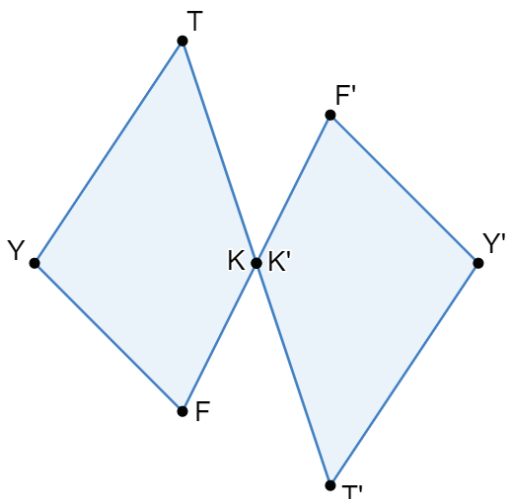
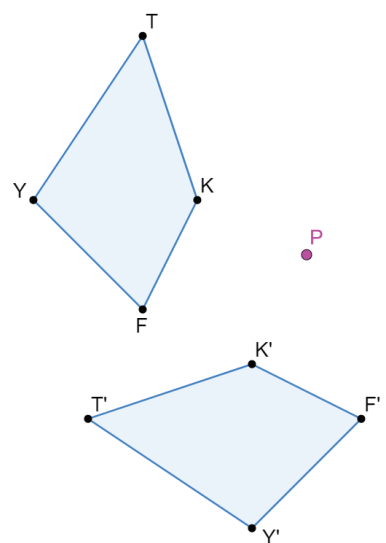
Transformation	Direction, Center, and Degree
	• Direction of rotation is _____. • Center of rotation is _____. • Degree of rotation is _____.
	• Direction of rotation is _____. • Center of rotation is _____. • Degree of rotation is _____.

Transformation	Direction, Center, and Degree
	- Direction of rotation is _____. - Center of rotation is _____. - Degree of rotation is _____.



1.5.4 Wrap-Up: Ticket Out the Door

- The table shows two different rotations of $TKFY$. Determine the direction of rotation, center of rotation, and the degree of rotation for each.

Transformation	Direction, Center, and Degree
	- Direction of rotation is _____. - Center of rotation is _____. - Degree of rotation is _____.
	- Direction of rotation is _____. - Center of rotation is _____. - Degree of rotation is _____.

Unit 1, Lesson 6: Transformations on the Plane – Part 2



Benchmark

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.



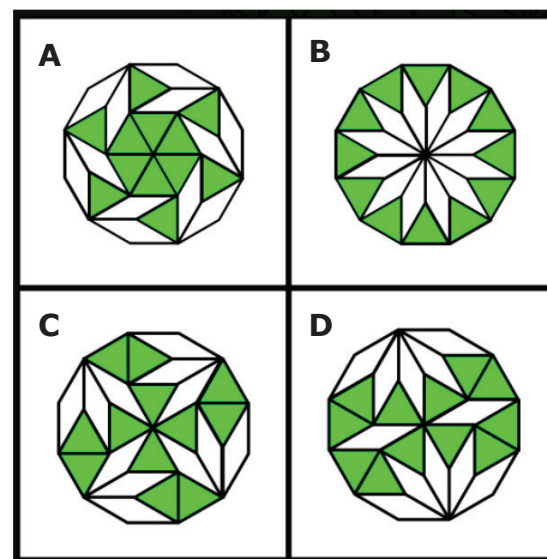
Learning Target

- ☐ I can describe a single transformation that was applied.



1.6.1 Warm-Up: Which One Doesn't Belong?

- The diagram shows four polygons.

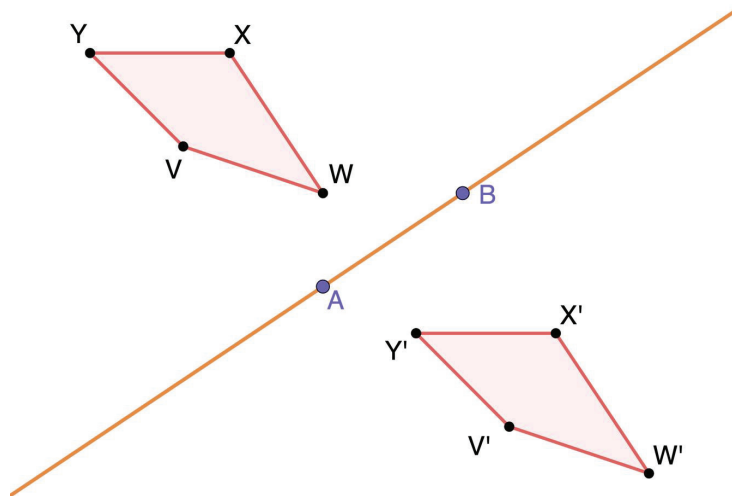


- Which polygon does not belong?
- Describe your reasoning.



1.6.2 Guided Instruction: Determining a Single Transformation

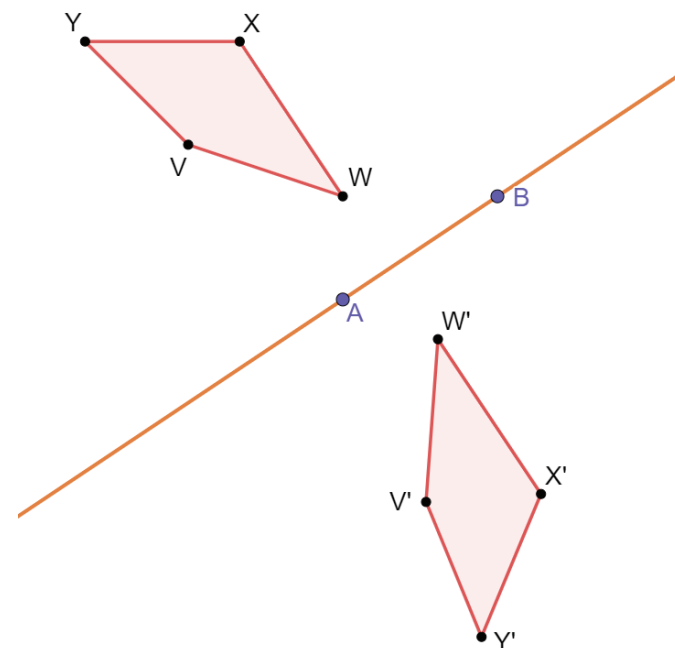
1. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.



Describe the single transformation that was used to create $V'W'X'Y'$.

Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	

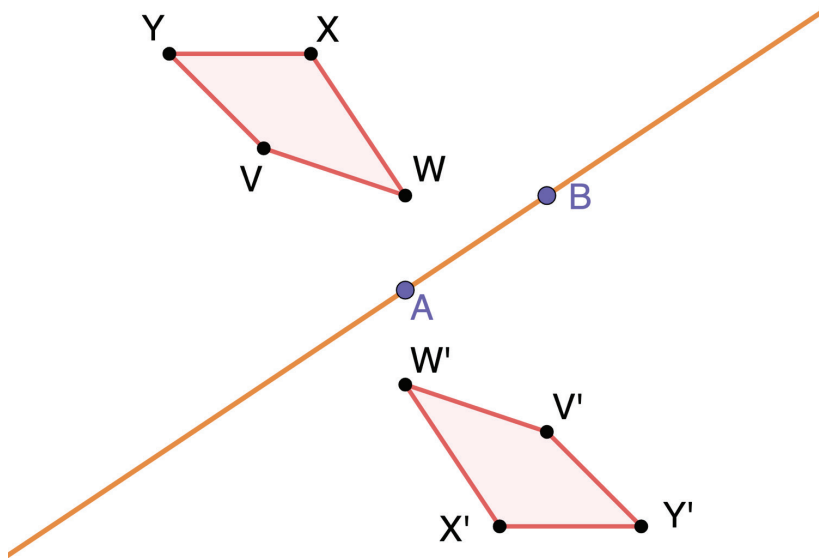
2. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.



Describe the single transformation that was used to create $V'W'X'Y'$.

Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	

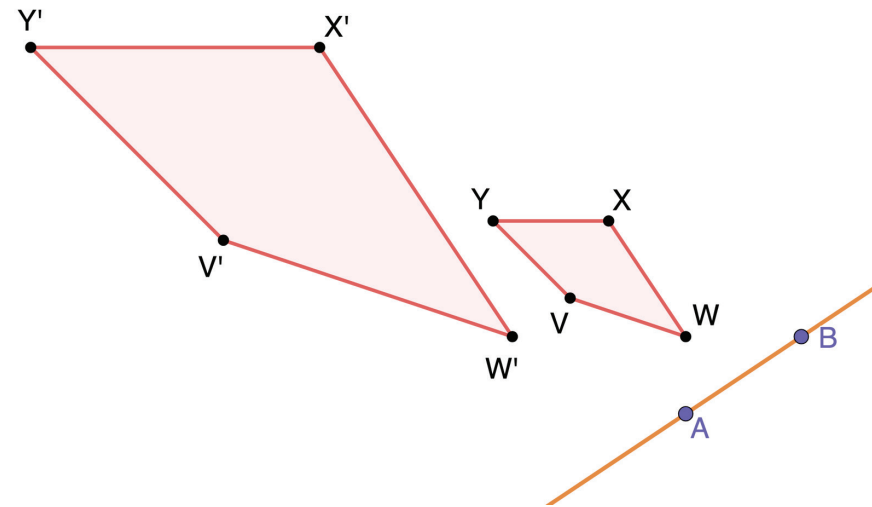
3. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.



Describe the single transformation that was used to create $V'W'X'Y'$.

Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	

4. The diagram shows $VWXY$ and $V'W'X'Y'$. $V'W'X'Y'$ is a transformation of $VWXY$.



Describe the single transformation that was used to create $V'W'X'Y'$.

Type of Transformation	Center, Line, or Movement	Degrees or Scale Factor
☐ Dilation ☐ Rotation ☐ Reflection ☐ Translation	☐ Center of dilation is _____. ☐ Center of rotation is _____. ☐ Line of reflection is _____. ☐ Down ☐ Left ☐ Right ☐ Up	

5. Complete the chart so it can be used to distinguish between similar-looking transformations. First, write your own hints. Then, exchange hints with your partner.

Transformation	My Hints	My Partner's Hints
Dilation		
Reflection		
Rotation		
Translation		

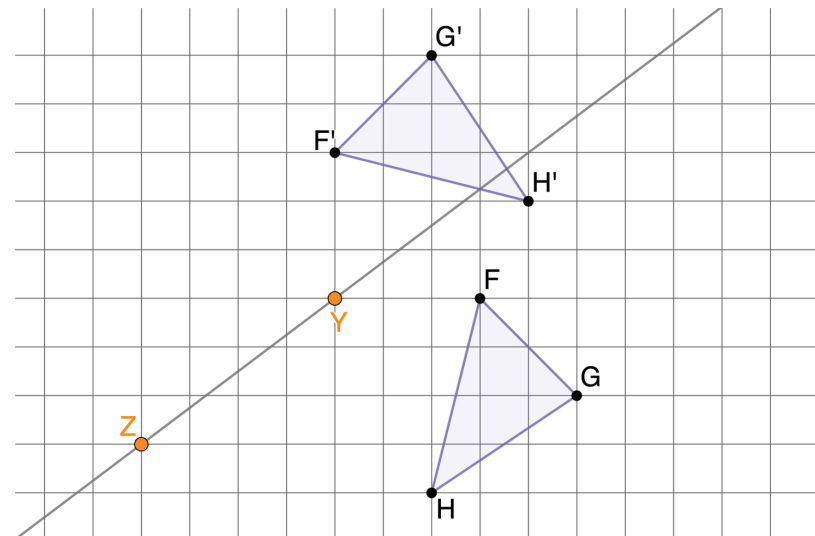
6. Which transformations resulted in congruent figures?
- ☐ Dilation
 - ☐ Reflection
 - ☐ Rotation
 - ☐ Translation
7. When a transformation results in a congruent figure, what is another name that can be used for that transformation?



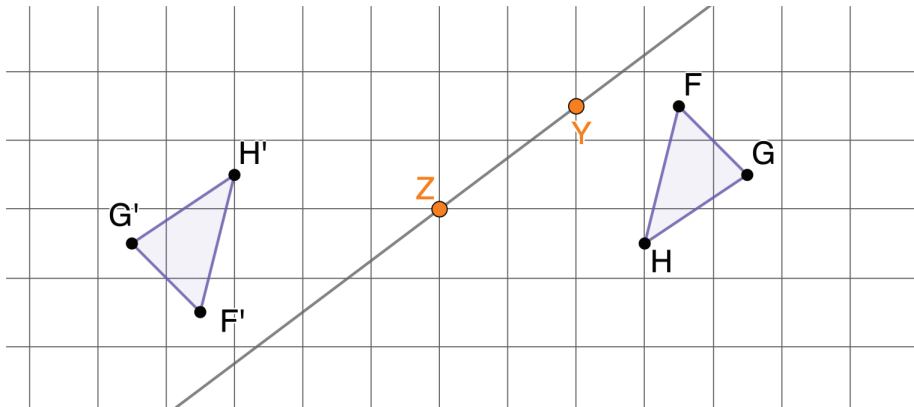
1.6.3 Guided Practice: Describing Transformations

Each of the diagrams shows $\triangle FGH$ and $\triangle F'G'H'$. $\triangle F'G'H'$ is a transformation of $\triangle FGH$.

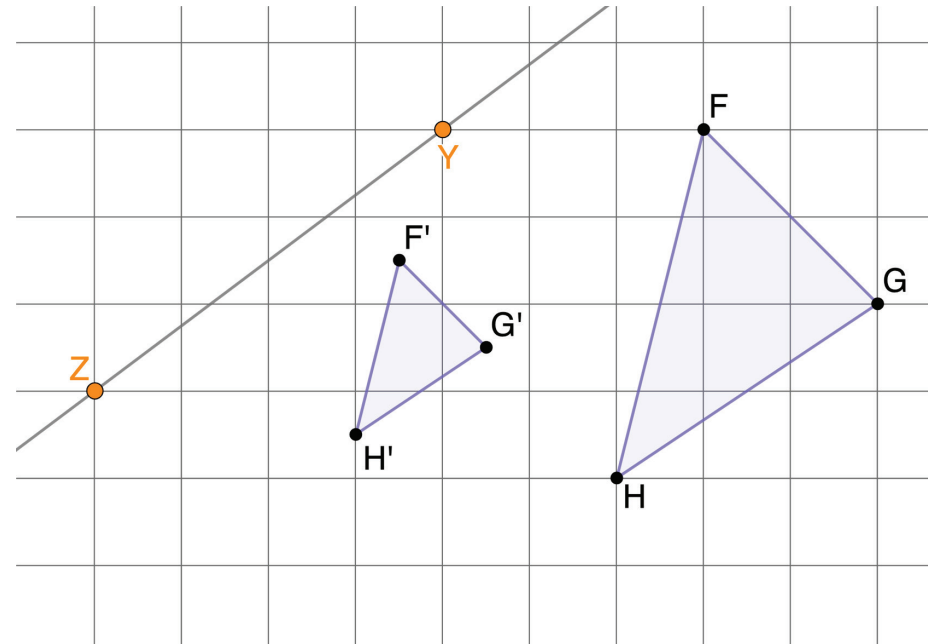
1. Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.



2. Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.



3. Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.

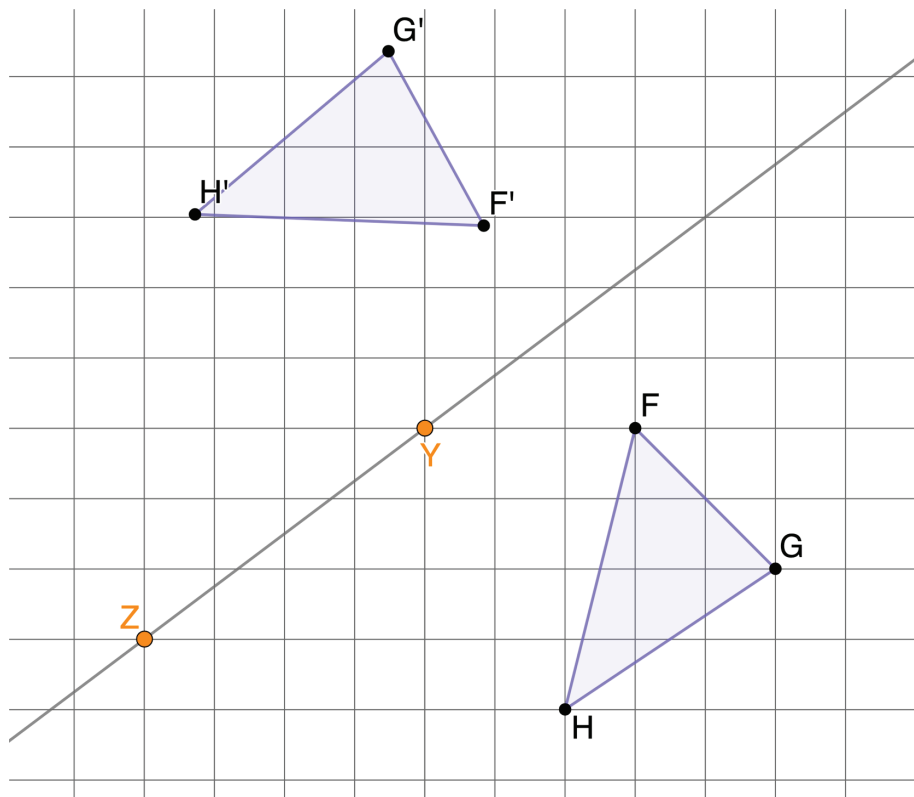


4. Review your hints chart and add anything else you think will be helpful.



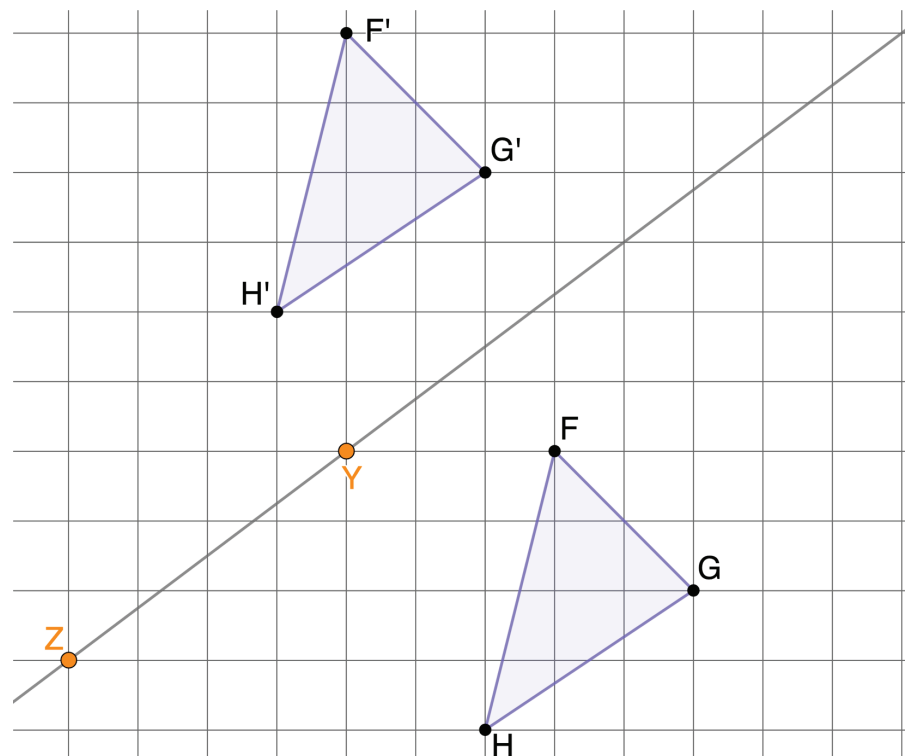
1.6.4 Your Turn!

1. The diagram shows $\triangle FGH$ and $\triangle F'G'H'$. $\triangle F'G'H'$ is a transformation of $\triangle FGH$.



Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.

2. The diagram shows $\triangle FGH$ and $\triangle F'G'H'$. $\triangle F'G'H'$ is a transformation of $\triangle FGH$.

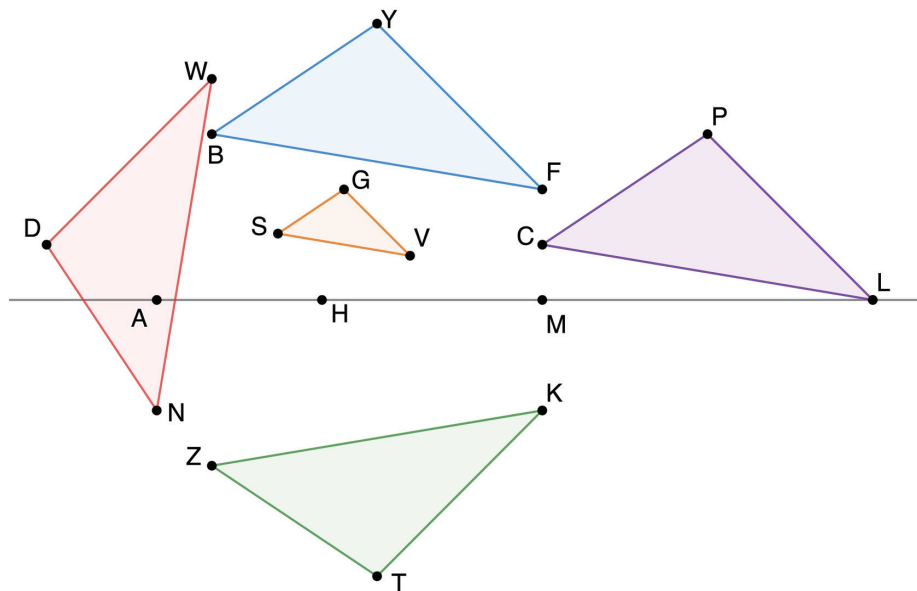


Describe the single transformation that was applied to $\triangle FGH$ to create $\triangle F'G'H'$.



1.6.5 Wrap-Up: Ticket Out the Door

1. The diagram shows four different single transformations of $\triangle YFB$.



Complete the chart by naming and describing the type of transformation that was applied to $\triangle YFB$ to result in each triangle.

Triangle	Type of Transformation	Description of the Transformation
$\triangle PLC$		
$\triangle TKZ$		
$\triangle GVS$		
$\triangle DWN$		

Unit 1, Lesson 7: Congruence and Similarity



Benchmarks

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.

MA.912.GR.2.8 Apply an appropriate transformation to map one figure onto another to justify that the two figures are similar.



Learning Targets

- ☐ I can recognize the relationship between corresponding sides and corresponding angles of congruent and similar figures.
- ☐ I can write congruence statements to relate corresponding parts following a rigid transformation.
- ☐ I can write similarity statements to relate corresponding parts following a dilation.



1.7.1 Warm-Up: 10,000 Hours to Awesome

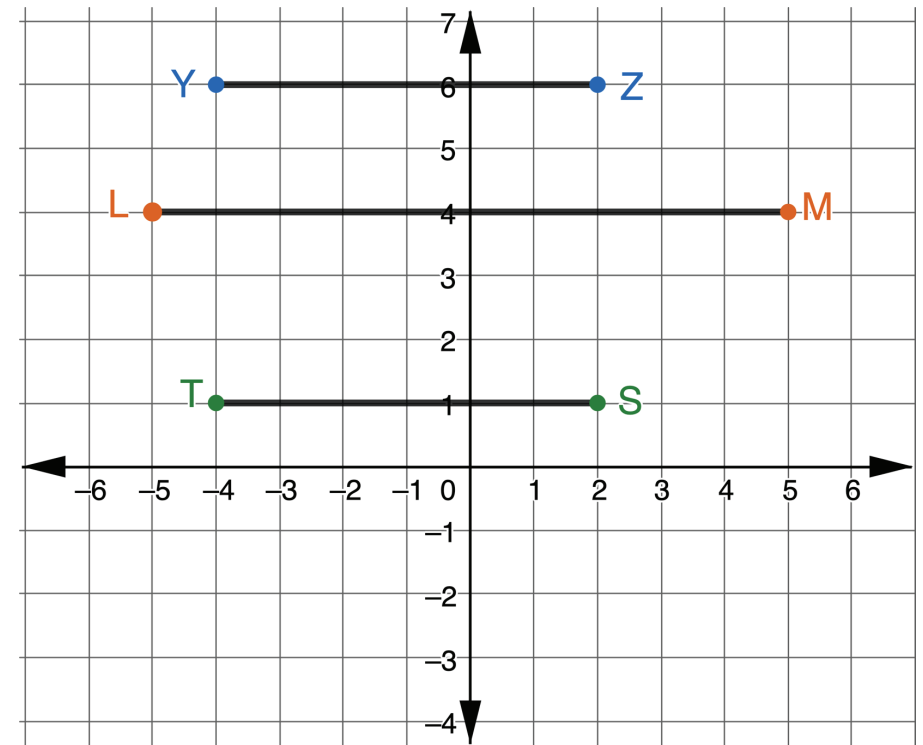
Best-selling author Malcolm Gladwell states in his book "Outliers" that it takes 10,000 hours of practice to be a phenom — so freakishly awesome or great at something. In 1973, Herbert Simon and William Chase wrote a paper published in *American Scientist*. Simon and Chase did experiments with chess-playing tasks and computer simulations to determine how long it would take someone to reach the grandmaster level in the game of chess. Their conclusion was that it could take as many as 50,000 hours staring at chess positions to become a chess grandmaster.

1. What skill, hobby, or activity would you like to become freakishly awesome at?
2. How many years is 10,000 hours?
3. How many years is 50,000 hours?



1.7.2 Collaboration: Line Segments

1. $\overline{LM}$, $\overline{TS}$, and $\overline{YZ}$ are shown on the coordinate grid.



- a. What do you notice about the three line segments?

- b. List at least two different ways to show that the length of $\overline{TS}$ is equal to the length of $\overline{YZ}$.
- c. What is the ratio of the length of $\overline{TS}$ to the length of $\overline{LM}$?
- d. What is the ratio of the length of $\overline{TS}$ to the length of $\overline{YZ}$?
- e. Describe a single transformation that could have been used to transform $\overline{TS}$ to create $\overline{LM}$.
- f. Draw a line that extends through the coordinate plane and passes through points L and T .
- g. Draw a line that extends through the coordinate plane and passes through points M and S .

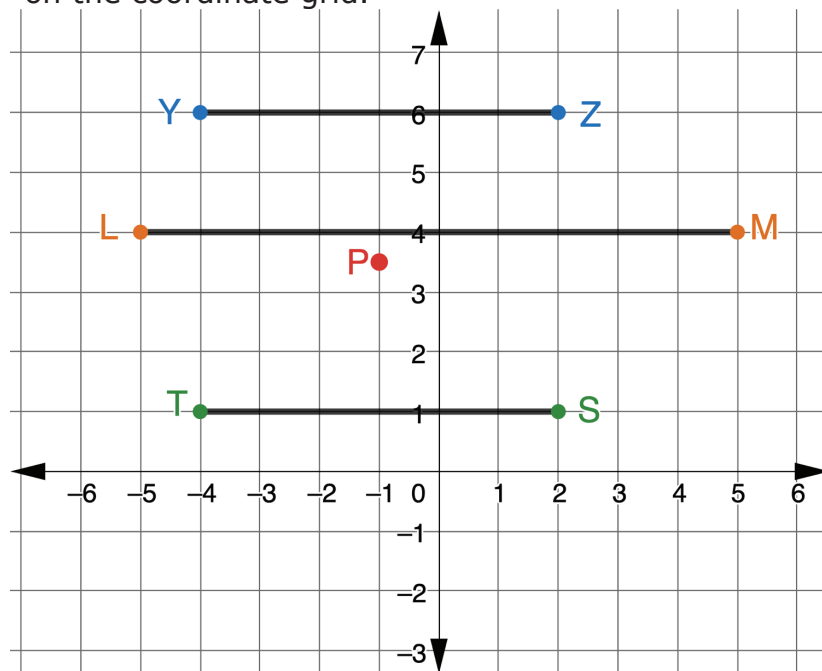
- h. Where do the two lines intersect?
- i. What is a single transformation that could have been used to transform $\overline{TS}$ to create $\overline{YZ}$?
- j. Ask two classmates the transformation they selected. Record their transformation and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Transformation	My Summary of Their Reason	Initials

2. Day'Ron and Zeb were discussing how to transform $\overline{TS}$ to create $\overline{YZ}$.

Day'Ron	Zeb
The only two single transformations are a translation of up 5, or a reflection across the line $y = 3.5$.	There are three single transformations. A translation of up 5, a reflection across the line $y = 3.5$, and a rotation of 180° .

Day'Ron disagreed with Zeb. To prove his point, he used a graphing utility to show $\overline{TS}$, $\overline{YZ}$, $\overline{LM}$ and point P on the coordinate grid.



Day'Ron said, "It is not possible to rotate $\overline{TS}$ 180° so that point T aligns with point Y and point S aligns with point Z ."

Zeb replied, "Yes, but that does not matter because the letters do not have to match exactly. If you rotate $\overline{TS}$ 180° about point P it will coincide with $\overline{YZ}$."

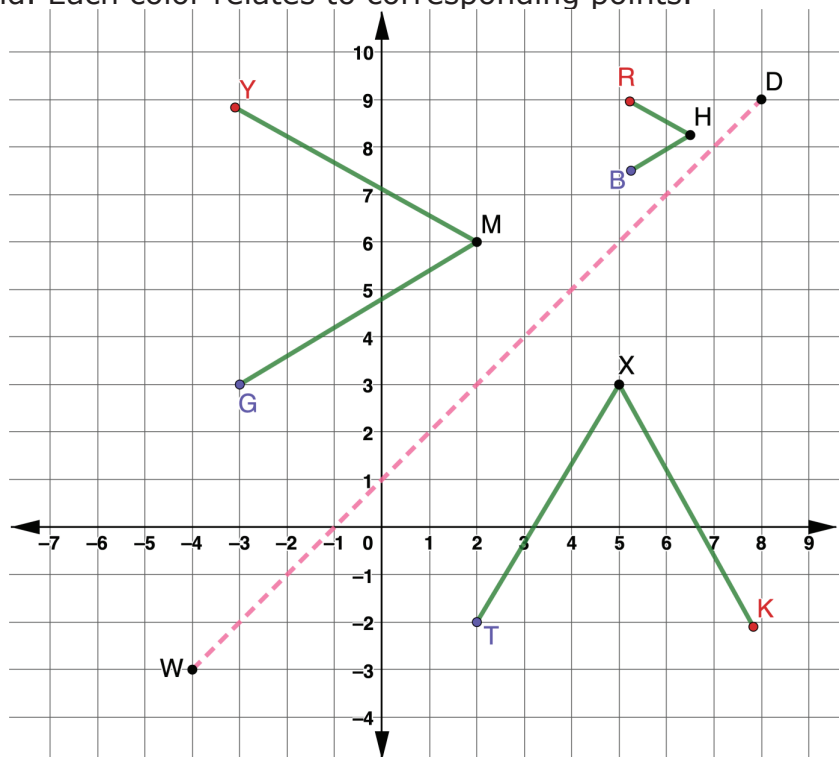
Discuss Day'Ron and Zeb's justifications with your partner. Summarize your discussion.

- Which transformation of $\overline{TS}$, $\overline{YZ}$, and $\overline{LM}$ showed congruence?
- Which transformation of $\overline{TS}$, $\overline{YZ}$, and $\overline{LM}$ showed similarity?



1.7.3 Guided Practice: Angles

$\overline{WD}$, $\angle YMG$, $\angle RHB$, and $\angle KXT$ are shown on the coordinate grid. Each color relates to corresponding points.



1. Describe a single transformation that could have been used to transform $\angle YMG$ to create $\angle KXT$.
2. Describe a single transformation that could have been used to transform $\angle YMG$ to create $\angle RHB$.

3. Use patty paper to create $\angle ALZ$ by rotating $\angle YMG$ 90° counterclockwise centered on point M .
4. Use patty paper to create $\angle CNF$ by translating $\angle YMG$ left 4 down 1.
5. Which angles have the same measure as $\angle YMG$?
6. Which transformation(s) resulted in an angle whose measurement is the same as $\angle YMG$?



1.7.4 Bring It Together: Congruence and Similarity

1. Complete the statements.
 - a. A rigid transformation results in a line segment that

is
is not

 congruent to the original line segment.
 - b. A dilation results in a line segment that

is
is not

 congruent to the original line segment.
 - c. A rigid transformation results in an angle that

is
is not

 congruent to the original angle.
 - d. A dilation results in an angle that

is
is not

 congruent to the original angle.



Two figures are **congruent** if they have exactly the same shape and size. Equivalently, two figures are congruent if one figure can be mapped to the other using a single rigid motion, or a sequence of transformations that includes only rigid motions.



Similarity is having exactly the same shape but not necessarily the same size. Two figures are similar if and only if one figure can be obtained from the other by a dilation, or a sequence of transformations that includes a dilation.

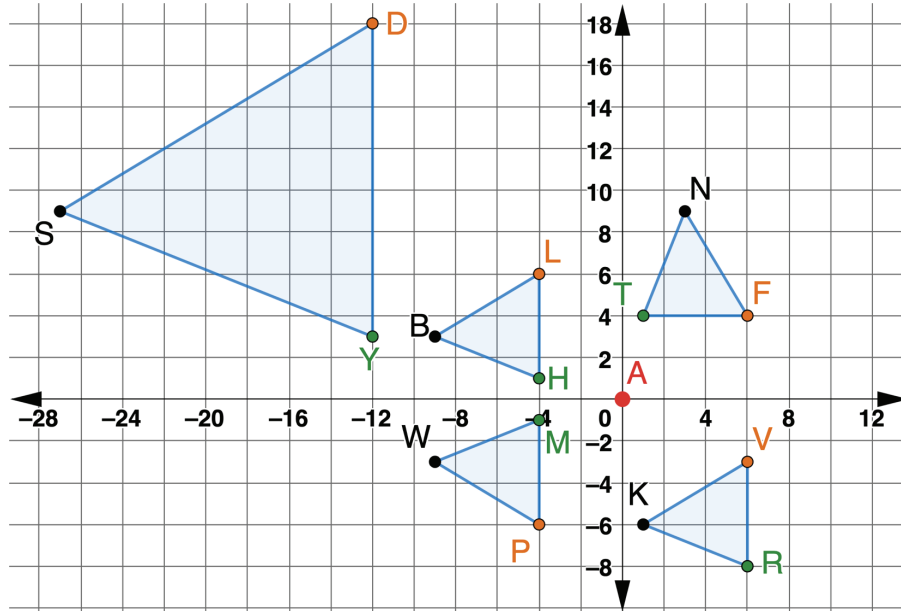
Two congruent figures can also be considered similar with a scale factor of 1. Although this is known, it is not often required to state that two congruent figures are also similar.

2. Which statement in Question 1 would be false for a dilation with a scale of factor of 1?



1.7.5 Guided Instruction: Congruence and Similarity Statements

1. Five triangles are shown on the coordinate grid. $\triangle FNT$, $\triangle VKR$, $\triangle PWM$, and $\triangle DSY$ are created using a single transformation of $\triangle LBH$. Each color relates to corresponding points.



- a. Complete the table.

Triangle	Description of Transformation	Congruent or Similar
$\triangle FNT$		
$\triangle VKR$		
$\triangle PWM$		
$\triangle DSY$		

- b. Use the information about the transformations to complete each statement.

$$\overline{LH} \cong \underline{\hspace{2cm}} \quad \overline{TN} \cong \underline{\hspace{2cm}} \quad \overline{YS} \sim \underline{\hspace{2cm}}$$

$$\angle DYS \cong \underline{\hspace{2cm}} \quad \overline{MP} \sim \underline{\hspace{2cm}} \quad \underline{\hspace{2cm}} \cong \overline{LH}$$

$$\overline{PW} \cong \underline{\hspace{2cm}} \quad \underline{\hspace{2cm}} \cong \angle LHB \quad \angle VKR \cong \underline{\hspace{2cm}}$$

- c. Discuss with your partner why there could be multiple answers to each statement. Summarize your discussion.



1.7.6 Wrap-Up: Cool Down

Complete the statements about today's lesson.

1. Something I learned today ...

2. I didn't know ...

3. One question I still have ...



Unit 1, Lesson H1: Symmetries



Benchmark

MA.912.GR.2.4 Determine symmetries of reflection, symmetries of rotation, and symmetries of translation of a geometric figure.



Learning Targets

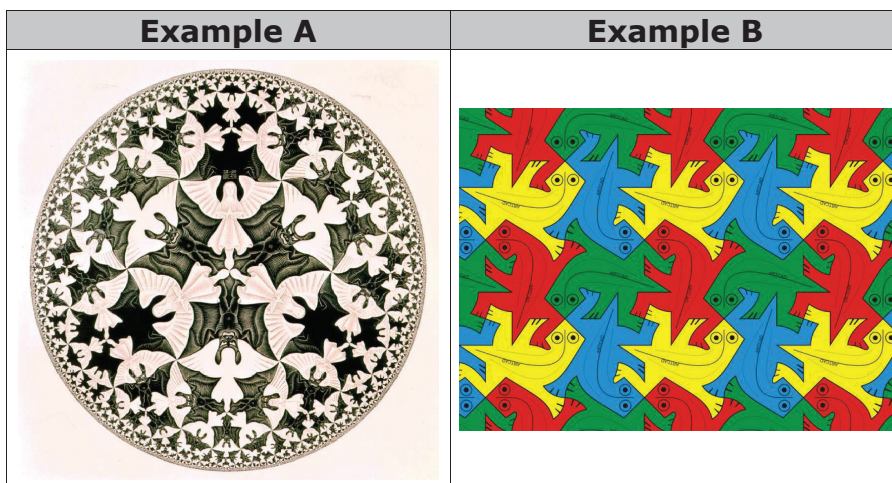
- ☐ I can determine symmetries of reflection.
- ☐ I can determine symmetries of rotation.
- ☐ I can determine symmetries of translation.



1.H1.1 Warm-Up: Artwork of M.C. Escher

M.C. Escher (1898-1972) was a Dutch artist whose work was inspired by mathematics. Though initially trained as an architect, he became one of the most famous and influential graphic artists in the world.

Below are two examples of Escher's work.



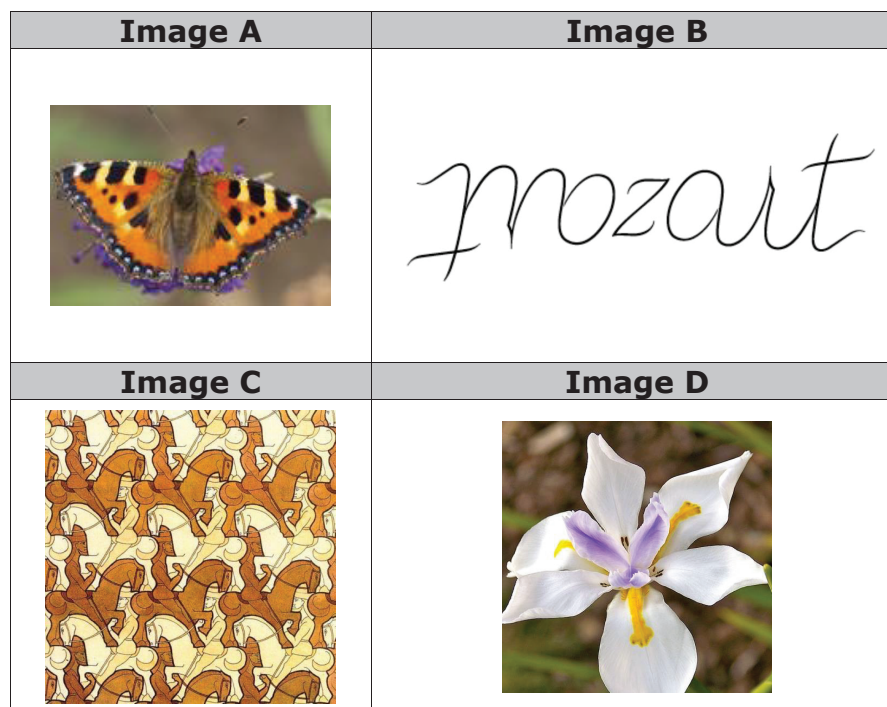
1. Describe any similarities and differences you notice about the two examples.

Similarities	Differences



1.H1.2 Exploration: Symmetry

The four images shown below have symmetry.



1. Discuss with your partner why each image has symmetry and which transformations could be used to show that one part of the image aligns with the other.

2. Complete the table. Some images may align with more than one type of transformation.

Image	Transformations
A	☐ reflection across a line ☐ rotation about a point ☐ translation
B	☐ reflection across a line ☐ rotation about a point ☐ translation
C	☐ reflection across a line ☐ rotation about a point ☐ translation
D	☐ reflection across a line ☐ rotation about a point ☐ translation

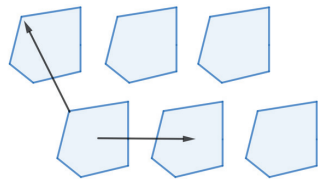
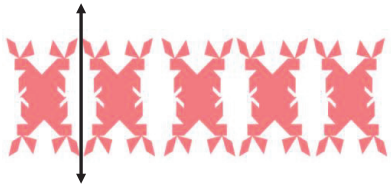
A **tessellation** is an arrangement of shapes closely fitted together in a repeated pattern without gaps or overlaps.

3. Which image is a tessellation?

Translational symmetry is when sliding a pattern is used to create symmetry.

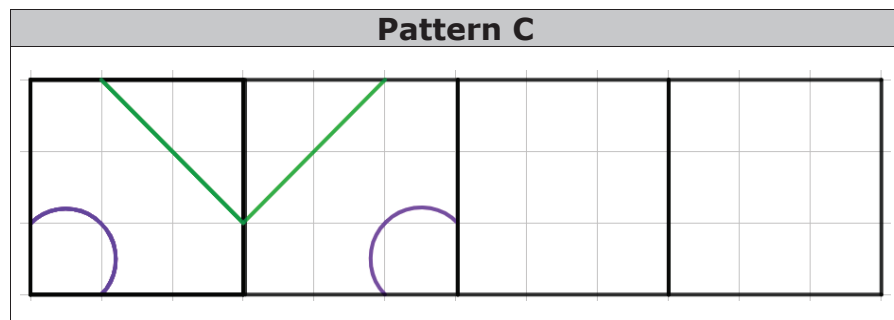
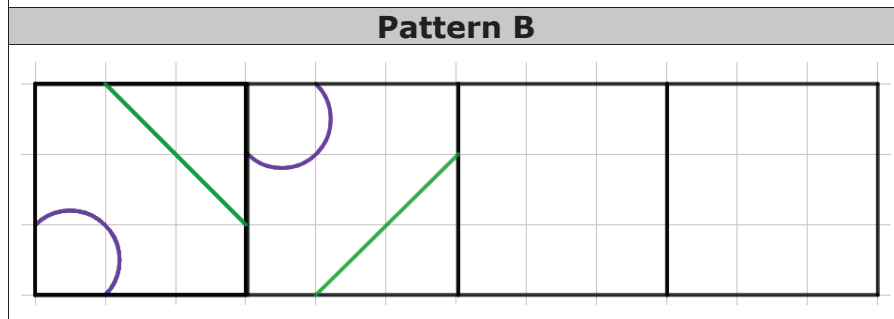
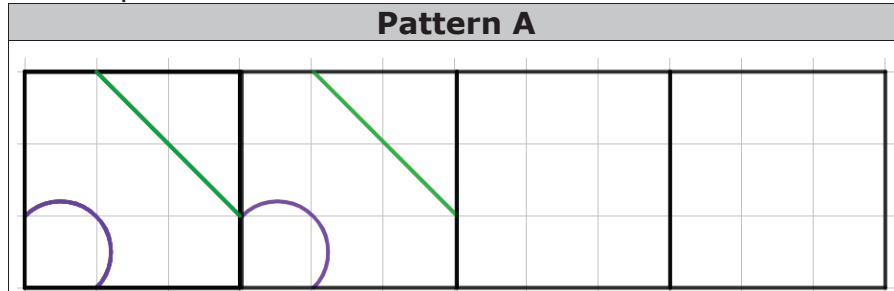
Reflectional symmetry is when a pattern can be reflected to create symmetry.

Rotational symmetry is when a rotation can be used to create symmetry. The table shows examples of each type of symmetry.

Image	Symmetry
	Translational symmetry
	Reflectional symmetry
	Rotational symmetry

4. Micah commented to his partner that he thought the picture for reflectional symmetry could also be translational symmetry. Discuss this with your partner.

5. For each pattern, use the first two squares to complete the pattern.



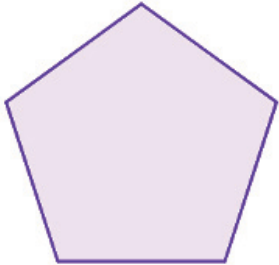
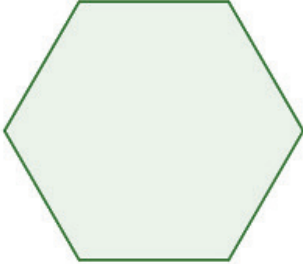
6. Work with your partner to determine the type of symmetry each pattern displays.

Pattern	Type of Symmetry
A	
B	
C	

The order of rotational symmetry of a shape is the number of times it can be rotated around a full circle and still look the same.

7. Complete the chart for each polygon.

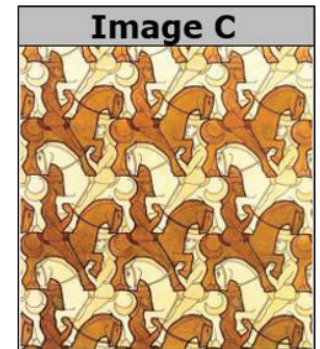
Polygon	Figure	Number of Lines of Symmetry	Order of Rotational Symmetry
Isosceles triangle			
Equilateral triangle			
Square			
Kite			

Polygon	Figure	Number of Lines of Symmetry	Order of Rotational Symmetry
Regular Pentagon			
Regular Hexagon			

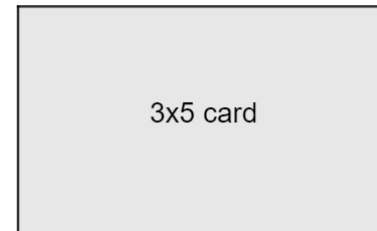


1.H1.3 Activity: Creating a Tessellation

Image C is a tessellation that was created by MC Escher, a famous Dutch artist.



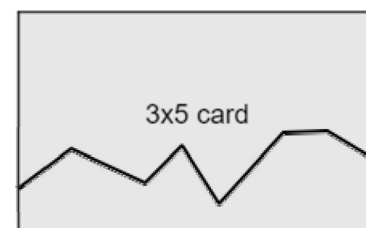
To create a tessellation, you will need a 3x5 index card and tape.



1. Create your own tessellation by choosing one of the following options. The more cuts you make, the more interesting your pattern will become.

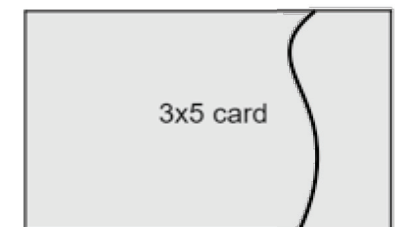
Horizontal Cut

Draw a curve or pattern along a horizontal side of the card.

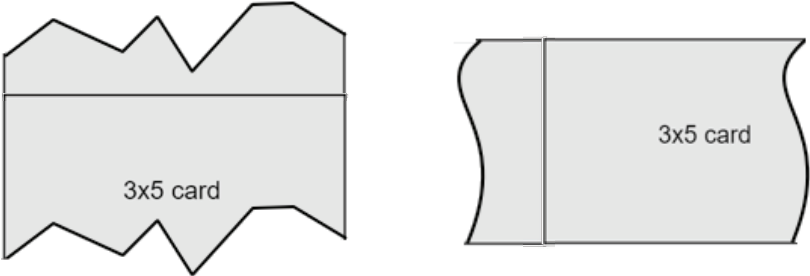


Vertical Cut

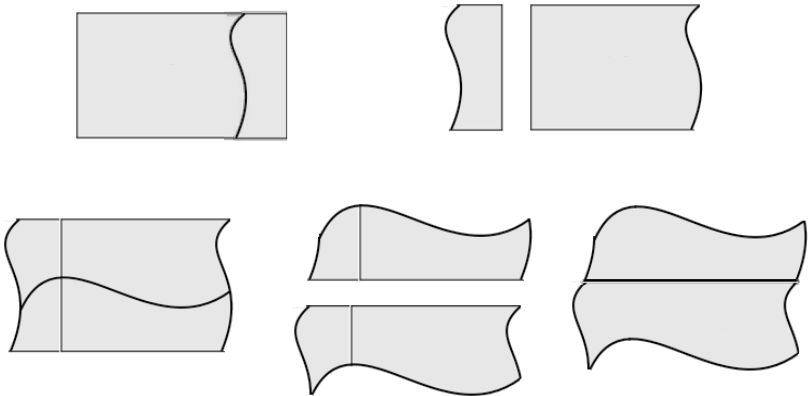
Draw a curve or pattern along a vertical side of the card.



Cut along the curve or pattern and shift the piece to the opposite side. Tape the piece on.



It is also possible to make multiple cuts as shown.



2. Use your card to create a tessellation by filling up a sheet of paper. Then, color your tessellation.



1.H1.4 Wrap-Up: Cool Down

1. Symmetry is often found in nature. Describe the transformations that show the symmetry of each object.

Image	Object	Transformation
	Each petal	
	Each arm of the starfish	



Unit 1, Lesson 8: Line Segments



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.GR.5.1 Construct a copy of a segment or an angle.

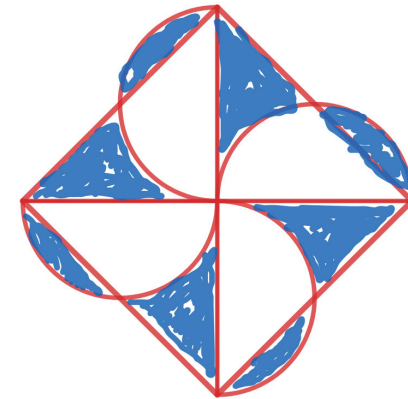


Learning Targets

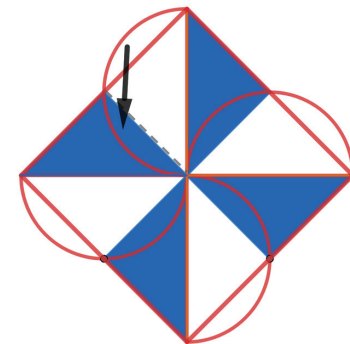
- ☐ I can construct a copy of a line segment.
- ☐ I can use the segment addition postulate.



1.8.1 Warm-Up: Semicircles and Triangles

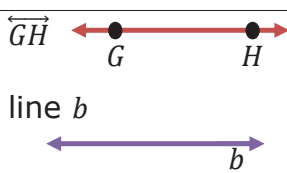
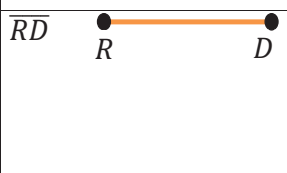
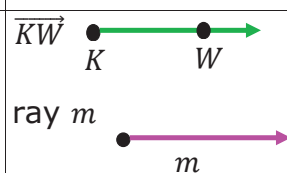


1. What do you notice about the picture?
2. If the radius of each semicircle is 5, then what is the length of each of the square's diagonals?
3. Dina determined the area of the shaded region using the drawing shown. Find the total shaded area.





1.8.2 Guided Instruction: Line Segments

Line	Line Segment	Ray
A line is an undefined term.	A part of a line that is bounded by two distinct end points.	A part of a line that has a fixed starting point and extends infinitely in one direction.
		

When naming a line segment, the line segment symbol should be used; however, when giving the length of a line segment, the symbol should not be used.

$\overline{TM}$

$TM = 12$ millimeters

- $\overline{RK}$ is shown. Use a ruler to measure the line segment in centimeters.



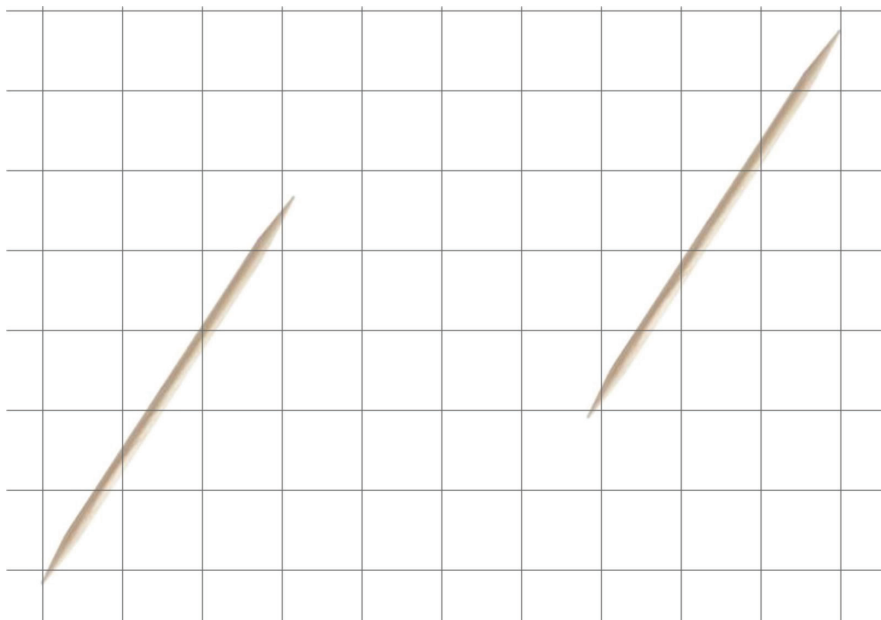
- Use patty paper to copy $\overline{LD}$. Then, draw line segment $\overline{XN}$ with the same length as $\overline{LD}$.



- Follow these steps to draw a copy of $\overline{LD}$ using a compass.

- Draw point C . This will be one endpoint of the new line segment.
- Set the compass' point on point L .
- Adjust the compass' width to point D so the compass' width is equal to the length of $\overline{LD}$.
- Without changing the compass' width, place the compass' point on C and draw an arc roughly where the other endpoint will be.
- Pick a point M on the arc that will be the other endpoint of the new line segment.
- Draw line segment $\overline{CM}$ using a straightedge.

4. Discuss with your partner how to show that the two toothpicks are congruent using each method in the table. Then, write a brief explanation for each method.



Transformation	
Patty Paper	
Compass	



Ruler Postulate

The points of a line can be placed in a one-to-one correspondence with the real numbers such that the distance between two distinct points is the absolute value of the difference of the corresponding real numbers.

5. $\overline{KC}$ and $\overline{YC}$ are shown.



- Discuss with your partner how aligning a ruler with $\overline{KC}$ illustrates the ruler postulate. Summarize your discussion.
- Use a ruler to measure each line segment. Include units.
 $KY = \underline{\hspace{2cm}}$ $YC = \underline{\hspace{2cm}}$ $KC = \underline{\hspace{2cm}}$
- Discuss with your partner why $KY + YC = KC$ should be true.

Points on the same line are called **collinear points**.



Segment Addition Postulate

If points A , X , and B are collinear points and point X is between points A and B , then $AX + XB = AB$.

6. $\overline{KW}$ and $\overline{WT}$ are shown, where $KW = 4x - 7$, $WT = 23.5$, and $KT = 7x - 3$.



- a. Determine the value of x .
- b. What is the length of $\overline{KW}$?



1.8.3 Guided Practice: Segment Addition

1. $\overline{YB}$, $\overline{BW}$, and $\overline{WH}$ are shown, where $YB = 2x$, $BW = x + 4$, $WH = 3x - 1$, and $BH = 35$.



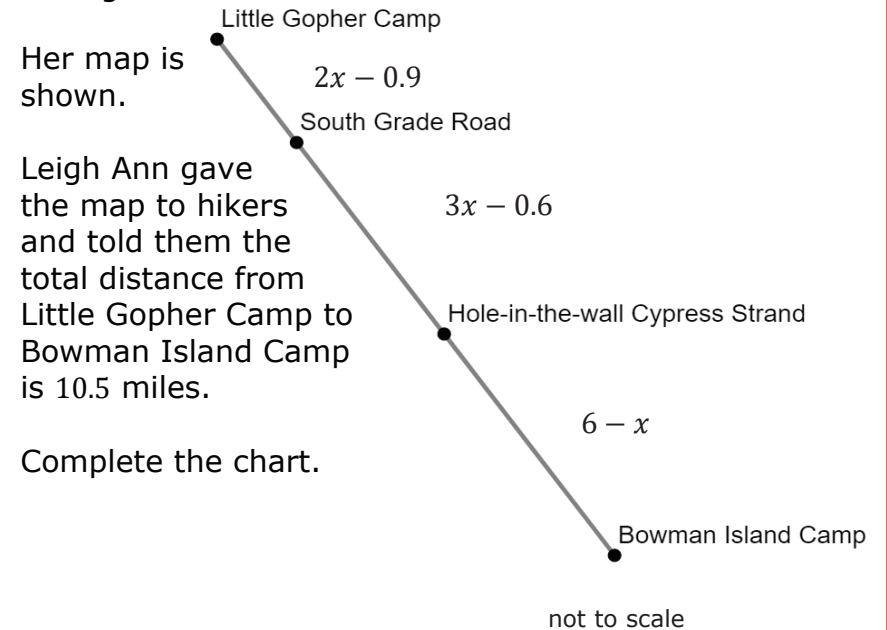
- a. Determine the value of x .
- b. What is the length of $\overline{YW}$?

2. Points K , L , M , and N are collinear such that $KM = LN$.
- Draw a diagram that represents points K , L , M , and N .
 - Compare your diagram with your partner. Discuss whether or not the diagrams drawn must all be the same. Summarize your discussion.
 - Determine the value of x if $KM = 6x - 6.2$ and $LN = 5x - 2$.
 - Using the information from c, determine the length of $\overline{KM}$.



1.8.4 Try It: Ocean to Lake Florida Trail

1. The Ocean to Lake Hiking Trail runs for approximately 63 miles, west from Hobe Sound Beach to Lake Okeechobee. Leigh Ann created a simple map of part of the trail that goes through J.W. Corbett Wildlife Management Area.



Hike	Distance (miles)
Little Gopher Camp to South Grade Road	
South Grade Road to Hole-in-the-wall Cypress Strand	
Hole-in-the-wall Cypress Strand to Bowman Island Camp	



1.8.5 Wrap-Up: Cool Down

1. Create an example of the Segment Addition Postulate and show the steps to finding the total length.

Unit 1, Lesson 9: Copying an Angle



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.GR.5.1 Construct a copy of a segment or an angle.



Learning Targets

- ☐ I can construct a copy of an angle.
- ☐ I can use the angle addition postulate.



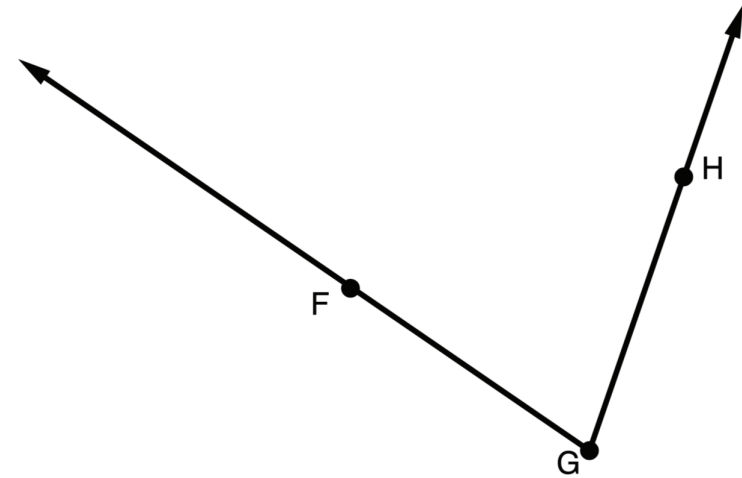
1.9.1 Warm-Up: Bell Work

1. Points C , H , and W are collinear. Point W is between C and H . $CW = 4x - 2$, $HW = 5x - 19$, and $CH = 7x + 5$.
 - a. Find the value of x .
 - b. What is the length of $\overline{HW}$?



1.9.2 Activity: Copying an Angle

1. $\angle FGH$ is shown.



- a. Discuss with your partner how to copy $\angle FGH$ using only a compass.

- b. Work with your partner to order the nine steps for copying $\angle FGH$. The first and last step are in the correct position.

1	Draw a point and label it X . Then, draw a ray starting at point X in any direction. Label the ray $\overrightarrow{XZ}$.
	Set the compass' point on point A and adjust the compass' width to point B .
	Without changing the compass' width, place the compass' point on Y and draw a similar arc to create point W where the two arcs meet.
	Set the compass' point on point G and adjust the compass' width to any width.
	Without changing the compass' width, place the compass' point on X and draw a similar arc. Label the point where the arc and $\overrightarrow{XZ}$ intersect point Y .
	Label the point where the arc crosses $\overrightarrow{GF}$ as point A .
	Label the point where the arc crosses $\overrightarrow{GH}$ as point B .
	Draw an arc across both rays of $\angle FGH$.
9	Draw $\overrightarrow{XW}$ using a straightedge.

- c. Compare your ordering with another group. If necessary, make adjustments to the order.
- d. Use your steps to draw $\angle WXY$, a copy of $\angle FGH$. If necessary, make adjustments to the order of your steps.



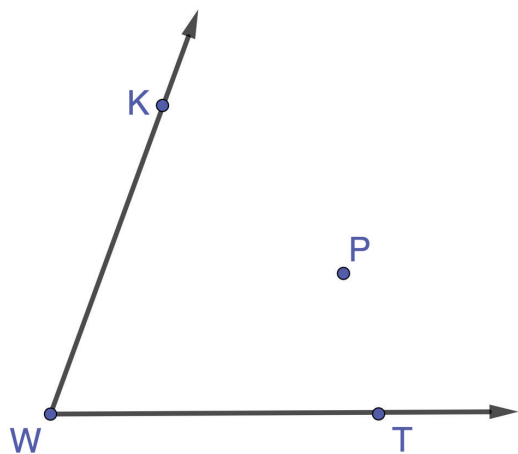
1.9.3 Guided Instruction: Angle Addition Postulate



Angle Addition Postulate

If point X is in the interior of $\angle AMB$, then
 $m\angle AMX + m\angle XMB = m\angle AMB$.

1. Point P lies in the interior of $\angle KWT$ as shown.



- a. Draw $\overline{WP}$.
- b. Use a protractor to measure each angle in degrees.

$m\angle KWT = \underline{\hspace{2cm}}$ $m\angle KWP = \underline{\hspace{2cm}}$ $m\angle PWT = \underline{\hspace{2cm}}$

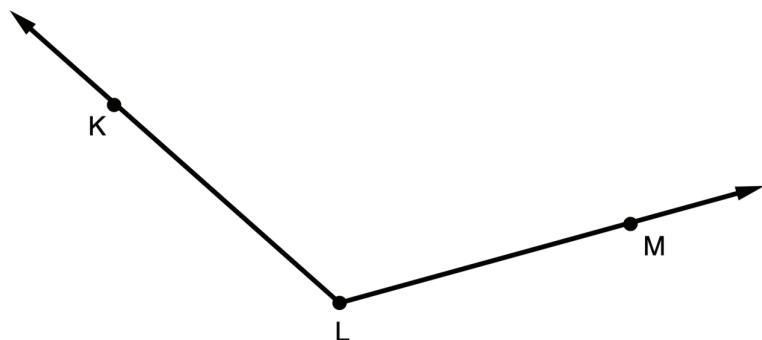
- c. Discuss with your partner why
 $m\angle KWP + m\angle PWT = m\angle KWT$ should be true.
Summarize your discussion.

- d. If $m\angle KWT = 70^\circ$, $m\angle KWP = (5x + 5)^\circ$ and
 $m\angle PWT = (4x - 7)^\circ$, determine the value of x .



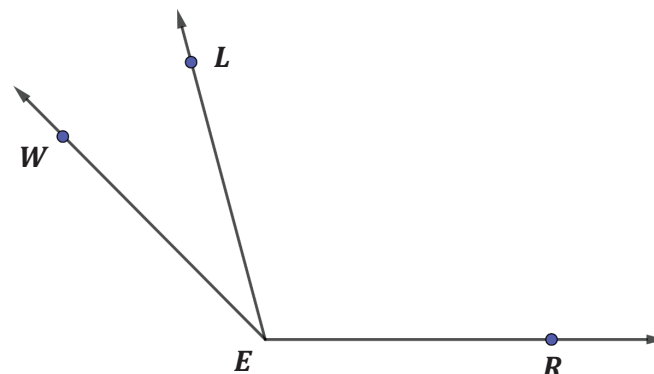
1.9.4 Try It: Copying an Angle

1. $\angle KLM$ is shown.



Draw $\angle RST$, a copy of $\angle KLM$.

2. Three angles are shown, $\angle WEL$, $\angle LER$, and $\angle WER$.



If $m\angle LER = (14.35x - 9.8)^\circ$, $m\angle WEL = (2.5x + 10)^\circ$, and $m\angle WER = 135^\circ$, what is $m\angle WEL$?



1.9.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about copying an angle.

Unit 1, Lesson 10: Segment Bisectors



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.



Learning Targets

- ☐ I can construct the bisector of a segment.
- ☐ I use the definition of a midpoint to solve problems.



1.10.1 Warm-Up: Notice and Wonder

In May 2021, the world's longest suspended pedestrian bridge opened in Arouca, Portugal. The bridge runs over 1700 feet across a canyon, nearly 600 feet above the Paiva River.



1. What do you notice?
2. What do you wonder?



1.10.2 Exploration: Segment Bisector

1. Complete the following.

- a. Copy $\overline{TK}$ onto patty paper.



- b. Follow the steps on the copy of $\overline{TK}$.
 - Fold the copy of $\overline{TK}$ that is on patty paper so the endpoints T and K align.
 - Place point M on $\overline{TK}$ where the paper folds.
 - Copy point M onto $\overline{TK}$ above.
 - Draw a line, a line segment, or a ray through point M .
- c. Discuss with your partner what you notice about $\overline{TM}$ and $\overline{MK}$.

The **midpoint** of a segment divides it into two congruent segments.

2. $\overline{WX}$ is shown where point P is the midpoint.



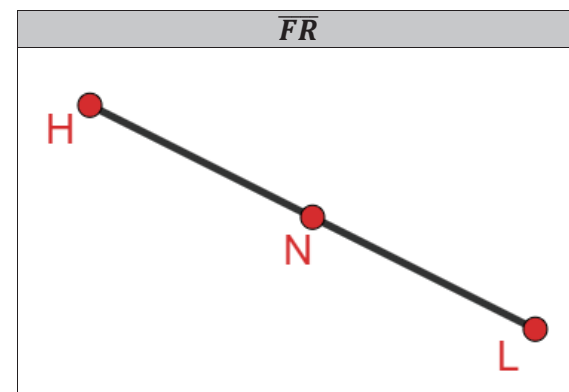
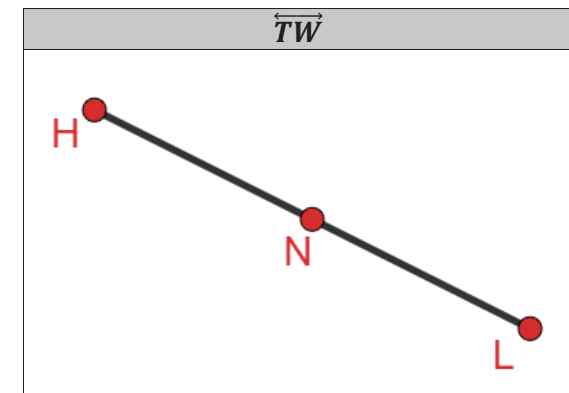
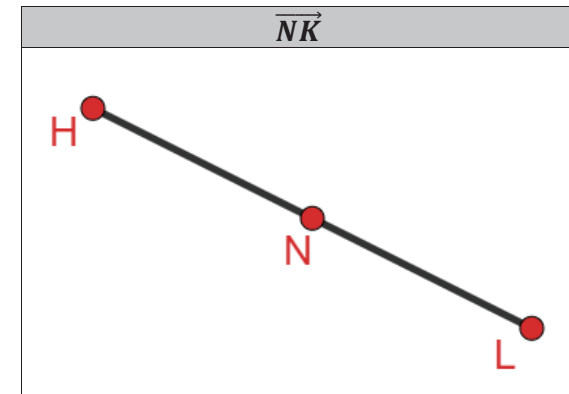
- a. Discuss with your partner what transformations can be used to show that $\overline{WP}$ maps to $\overline{PX}$.
- b. Ask two classmates which transformations they selected. Record their transformation and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Transformation	My Summary of Their Reason	Initials

A **segment bisector** is a line, ray, or line segment that cuts a line segment into two equal parts.

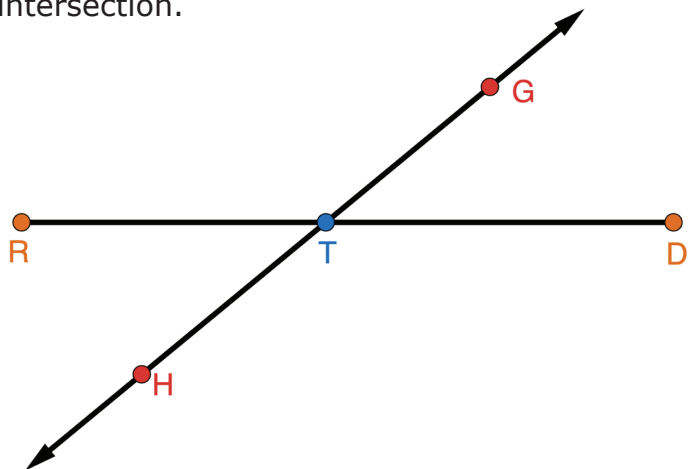
3. $\overline{HL}$ is shown where point N is the midpoint.

- a. Use each description to draw a segment bisector for $\overline{HL}$.



- b. Compare your drawings with your partner.
- c. Using the definition of a segment bisector, discuss with your partner if all of the segment bisectors have to be at the same angle with $\overline{HL}$. Summarize your discussion.

4. $\overline{RD}$ and $\overleftrightarrow{HG}$ are shown where point T is the point of intersection.



- a. Use patty paper to copy $\overline{RD}$.
- b. Fold the copy of $\overline{RD}$ to determine if point T is a midpoint.
- c. Complete the sentence.
 $\overleftrightarrow{HG}$ ☐ is ☐ is not a segment bisector because point T
☐ is ☐ is not a midpoint.



1.10.3 Guided Instruction: Determining Lengths

1. $\overleftrightarrow{KH}$ bisects $\overline{GB}$ at point K .
 - a. Draw a diagram of the description.
 - b. Determine the value of x if $GK = 7x + 36$ and $KB = 13x - 18$.
 - c. What is the length of $\overline{GB}$?

2. $\overline{YH}$ and $\overline{MV}$ bisect each other at point W .

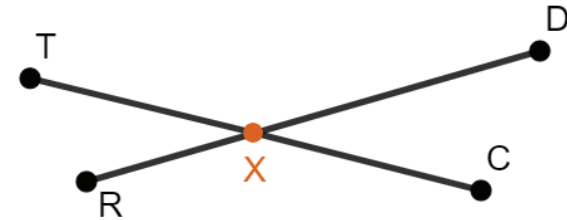
a. Draw a diagram of the description.

b. Determine the value of b if $YH = 12b - 4.5$, $MV = 17b + 0.75$, and $MW - WH = 13.25$.



1.10.4 Guided Practice: Determining Lengths

1. $\overline{RD}$ bisects $\overline{TC}$ at point X .



a. Determine the value of x if $TX = \frac{16}{3}x - 41$ and $XC = 3x + 22$.

b. What is the length of $\overline{TC}$?



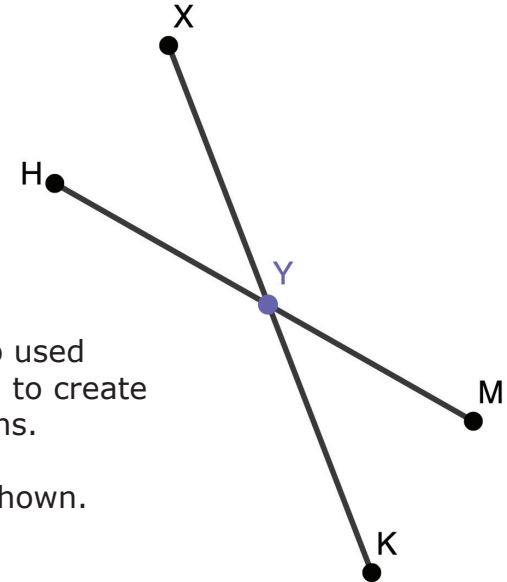
1.10.5 Your Turn!

1. $\overleftrightarrow{BL}$ bisects $\overline{HY}$ at point C .
 - a. Draw a diagram of the description.
 - b. Determine the value of x if $HC = 5x + 9$ and $CY = 8x - 36$.
 - c. What is the length of $\overline{HY}$?



1.10.6 Wrap-Up: Error Analysis

1. $\overline{XK}$ and $\overline{HM}$ bisect each other at point Y , where $XY = 5x + 3$, $HY = 2y + 5$, $XK = 11x + 2$, and $YM = 3y - 2$.



Taliyah and Giancarlo used the given information to create two different equations.

Their equations are shown.

Taliyah	Giancarlo
$5x + 3 = 2y + 5$	$5x + 3 = 2(11x + 2)$

Both Taliyah and Giancarlo made a mistake. If Taliyah was trying to find the value of y and Giancarlo was trying to find the value of x , determine each of their errors. Explain each error.

Taliyah	Giancarlo

Unit 1, Lesson 11: Perpendicular Bisectors



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.



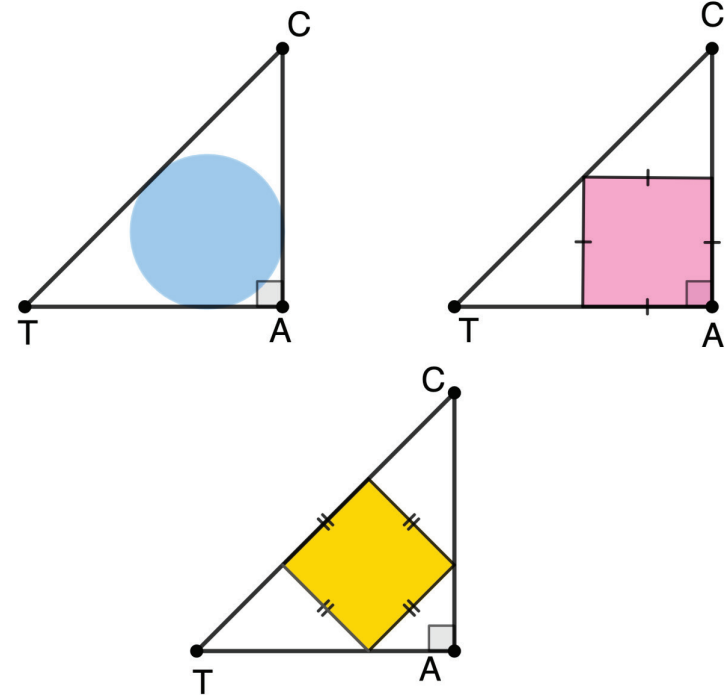
Learning Targets

- ☐ I can construct a perpendicular bisector.
- ☐ I can use rigid transformations to show that any triangle formed by drawing two segments from the endpoints of a segment to a point on its perpendicular bisector is isosceles.
- ☐ I can use the properties of a perpendicular bisector to solve problems.



1.11.1 Warm-Up: Compare Areas

$\triangle CAT$ is an isosceles right triangle.



1. Which inscribed figure has the greatest area? Justify your choice.



1.11.2 Exploration: Constructing Perpendicular Bisectors

1. Use the steps shown to construct the perpendicular bisector of $\overline{AB}$.

Step 1: Set the compass point at point A and open the compass to a width approximately two-thirds the length of $\overline{AB}$.

Step 2: Without changing the width of the compass, draw an arc above and below the line centered at point A .

Step 3: Repeat Steps 1 and 2 to create an arc centered at point B .

Step 4: Mark the two points of intersections of the arcs as C and D . Draw a line creating $\overline{CD}$ that intersects $\overline{AB}$.

Step 5: Label the point of intersection for $\overline{CD}$ and $\overline{AB}$ as point X .



2. Use a protractor and ruler to find each of the measurements.

Angle	Measurement	Segment	Measurement
$\angle AXD$		$\overline{AB}$	
		$\overline{AX}$	
$\angle AXC$		$\overline{BX}$	
		$\overline{CB}$	
$\angle BXD$		$\overline{DB}$	
		$\overline{CA}$	
$\angle BXC$		$\overline{DA}$	

3. Complete the statements.

The angle formed by the line of intersection of $\overline{AB}$ and

$\overline{CD}$ is
☐ 85°
☐ 90°
☐ 105°
 . The perpendicular bisector $\overline{CD}$ divides $\overline{AB}$ into
☐ 2
☐ 3
☐ 4
 congruent parts.

4. On the perpendicular bisector $\overline{CD}$, add three points labeled Q , R , and S in different places.

5. Using a ruler, measure the length of each line segment.

Segment	Length
$\overline{QA}$	
$\overline{QB}$	
$\overline{RA}$	
$\overline{RB}$	
$\overline{SA}$	
$\overline{SB}$	

6. Write a conjecture about the endpoints of a line segment and any point on its perpendicular bisector.

7. Share your conjecture with a partner. Summarize your partner's conjecture and have them initial that your summary is correct.

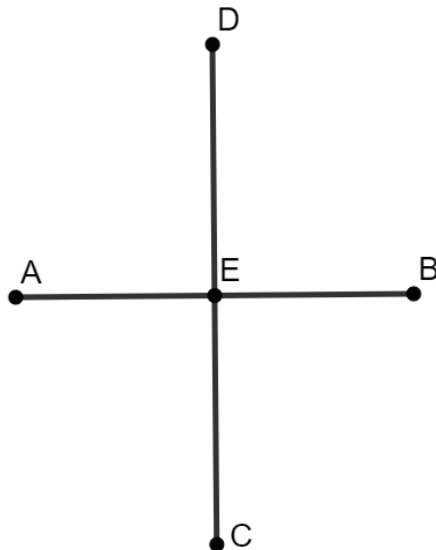
Partner's Conjecture	Partner's Initials





1.11.3 Activity: Perpendicular Bisectors and Triangles

- Copy $\overline{AB}$ and its perpendicular bisector $\overline{CD}$ on patty paper.



- Draw $\overline{AD}$ and $\overline{BD}$.
- Make a conjecture about $\triangle ADE$ and $\triangle BDE$.



A **right triangle** is a triangle containing an interior right angle.

- Reflect $\triangle ADE$ over $\overline{CD}$.
- Discuss with your partner what the reflection shows is true about $\triangle ADE$ and $\triangle BDE$. Summarize your discussion.



An **isosceles triangle** contains at least two equal length sides and two equal interior angle measures.

- Complete the statements.

In $\triangle ABD$, $\angle DAB$ is congruent to

☐ $\angle ADB$
☐ $\angle DBA$

 because a

reflection is a rigid motion that preserves angle measure.

Since $m\angle DAB$ is equal to

☐ $m\angle ADB$,
☐ $m\angle DBA$,

 and

☐ $AD = AB$,
☐ $AD = BD$,
☐ $AE = BE$,
☐ None of these

by definition $\triangle ABD$ is a(n)

☐ scalene
☐ isosceles
☐ equilateral

 triangle.



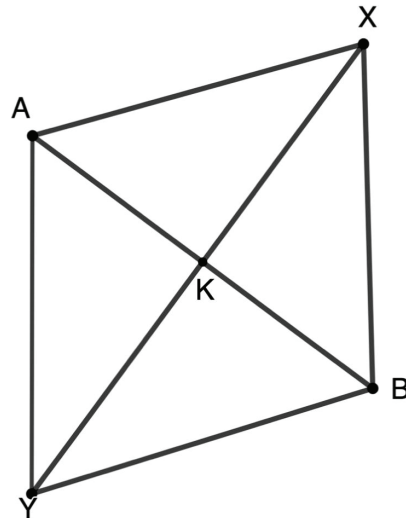


1.11.4 Bring It Together: Perpendicular Bisectors and Triangles



If a point is on the perpendicular bisector of a line segment, then it is equidistant from the endpoints of the line segment.

In the diagram, $\overline{XY}$ is the perpendicular bisector of $\overline{AB}$, $XA=7.7$ millimeters (mm), $YB=6.85$ mm, and $AB=7.7$ mm. The figure is not drawn to scale.



1. Label all segment measurements from the given information on the image. Then, label all other segments based on the relationship of a line segment and its perpendicular bisector.
2. Given $\overline{XY}$ is the perpendicular bisector of $\overline{AB}$, label all of the congruent angles.

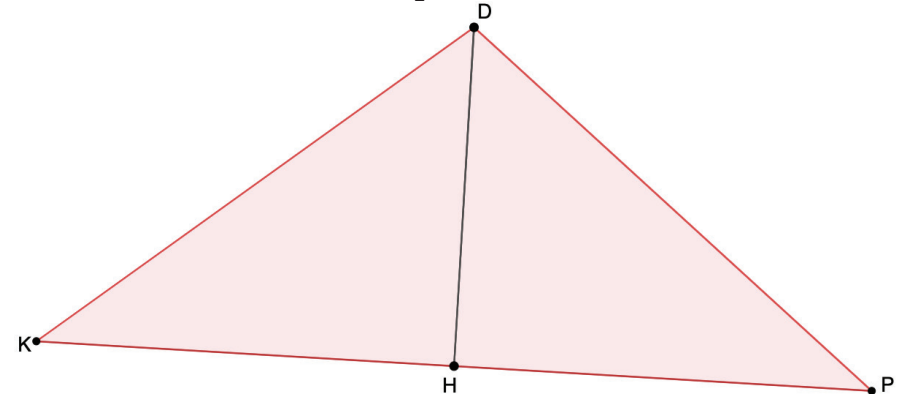


The triangle formed by a line segment and two segments drawn from the endpoints of the segment to a point on the perpendicular bisector of the segment is an isosceles triangle.



1.11.5 Guided Practice: Perpendicular Bisectors

1. $\overline{DH}$ is the perpendicular bisector of $\overline{KP}$, where $KD = -x - 1$ and $PD = -\frac{x}{2} + 19$.

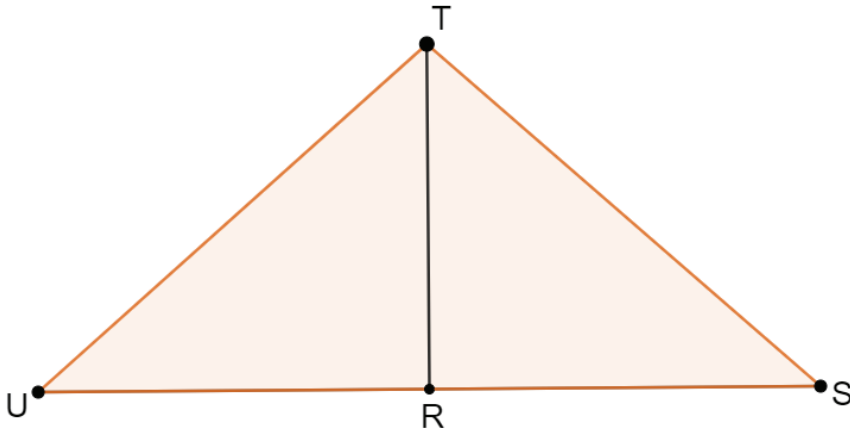


- a. What is the value of x ?
- b. What is the length of $\overline{KD}$?

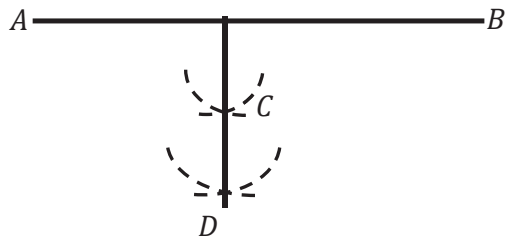


1.11.6 Your Turn!

1. $\overline{RT}$ is the perpendicular bisector of $\overline{US}$. $UT = 8y - 12$ and $ST = 3y + 23$. Find the value of y .



2. Describe the error that was made in the construction of a perpendicular bisector.



1.11.7 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

Unit 1, Lesson 12: Angle Bisectors



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.



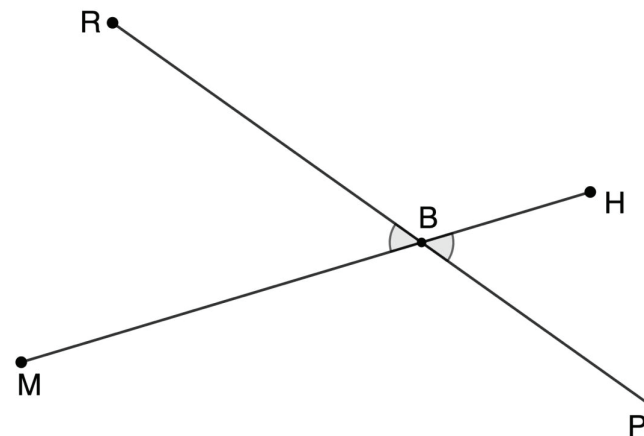
Learning Target

- ☐ I can construct an angle bisector.



1.12.1 Warm-Up: Agree or Disagree

1. Line segments MH and RP intersect at point B .



- a. Jamessia says $\angle MBR$ and $\angle PBH$ have the same measure. Malik disagrees and says that obviously the measure of $\angle MBR$ is larger.

Where do you stand?

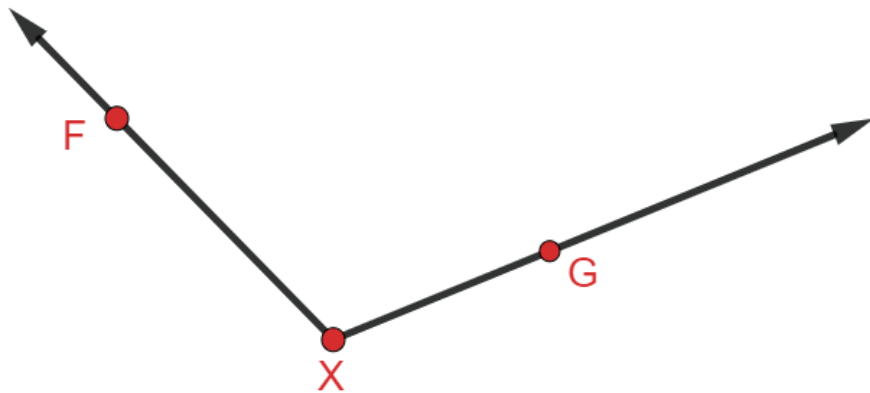
- ☐ I agree with Jamessia.
- ☐ I agree with Malik.
- ☐ I disagree with both students.

- b. Explain your choice for Part A.



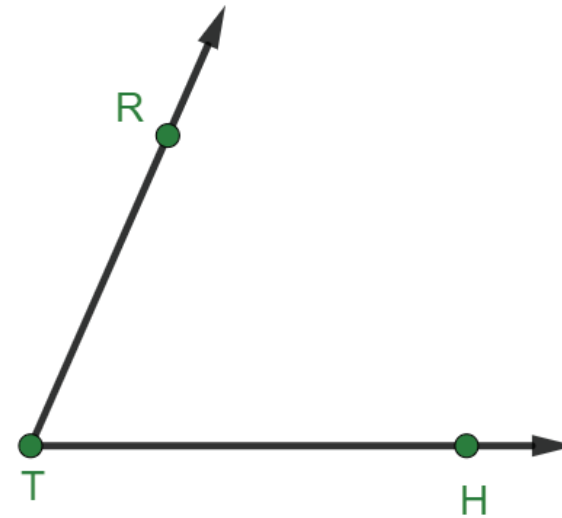
1.12.2 Collaboration: Bisecting an Angle

1. $\angle FXG$ is shown.



- Measure the angle with a protractor.
- Draw $\overrightarrow{XY}$ that bisects $\angle FXG$.
- Discuss with your partner why bisecting $\angle FXG$ using a protractor and a ruler may not be accurate. Summarize your discussion.

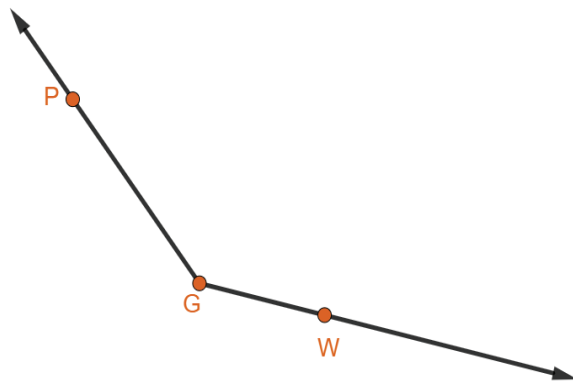
2. $\angle RTH$ is shown.



- Use patty paper to copy $\angle RTH$.
- Fold the patty paper so that $\overrightarrow{TR}$ and $\overrightarrow{TH}$ lie on each other. Open the paper and draw $\overrightarrow{TM}$ along the crease.
- Explain how folding the patty paper bisects the angle.

Another way to more accurately bisect an angle is to use a compass and a ruler.

3. $\angle PGW$ is shown.



Work with your partner and use the steps below to bisect $\angle PGW$.

- Place the compass' point on point G .
- Adjust the compass' width to any width that will stay on the angle.
- Use the setting to draw an arc that extends across each leg of $\angle PGW$. Label the intersections of the arc and the two sides of the angle points F and H .
- Place the compass' point on point F . Keep the width of the compass the same or open the compass wider than the width used to draw the arc in the previous step. Draw an arc in the interior of the angle.
- Without changing the compass' setting, place the compass' point on point H and make an arc in the interior of the angle. The two small arcs in the interior of $\angle PGW$ should be intersecting. If they are not intersecting, adjust the compass to a larger width and draw new arcs from points F and H .
- Draw and label point Y where the two arcs intersect.
- Use a straightedge to draw a ray from point G through point Y .
- $\overrightarrow{GY}$ bisects $\angle PGW$.

4. Olivia found some steps for bisecting an angle on the internet. She wrote them down and then mixed them up to test herself. Her mixed-up steps are shown. Work with your partner to order the steps 1-9.

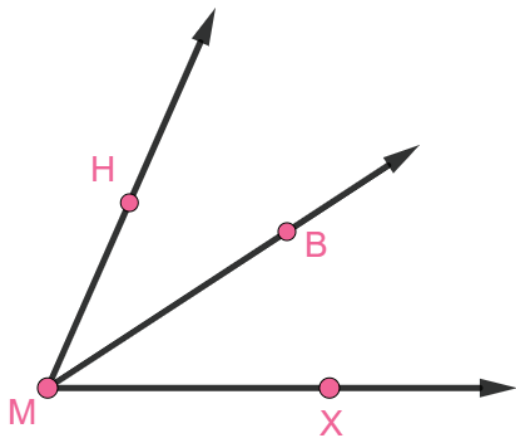
	Draw a ray from the vertex of the angle to the intersection of the two small arcs.
	Stretch the compass to any length that stays on the angle.
	Without changing the span on the compass, place the point of the compass on the other point made by the arc and the other side of the angle and make a similar arc.
	Place the compass point on one of the two new points made by the arc.
	Change the compass width so that the pencil stays between the rays of the angle.
	Place compass point on the vertex of the angle.
	Swing the compass to make an arc that crosses both rays of the angle.
	Draw an arc in the interior of the angle.
	Mark the point where the two small arcs in the interior of the angle intersect.



1.12.3 Guided Instruction: Using Angle Bisectors to Solve Problems

An **angle bisector** is a line, ray, or line segment that cuts an angle into two equal parts.

1. $\angle HMX$ is shown where $\overline{MB}$ bisects $\angle HMX$.



If $m\angle HMX = (7x + 3)^\circ$ and $m\angle BMX = (3x + 6)^\circ$, then find the value of x .

2. $\overline{DY}$ bisects $\angle WDN$ such that $m\angle YDN = (10x - 4)^\circ$ and $m\angle WDN = (17x + 16.75)^\circ$.

- a. Draw a diagram of the description.

- b. Yordis and Hae-Won used the given information to determine the value of x . Their equations are shown.

Yordis	Hae-Won
$10x - 4 = \frac{1}{2}(17x + 16.75)$	$\frac{1}{2}(10x - 4) = 17x + 16.75$

- c. Discuss with your partner which student made a mistake setting up the equations.
- d. Write a short note to the student who made a mistake in setting up the equations to explain their error to them.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

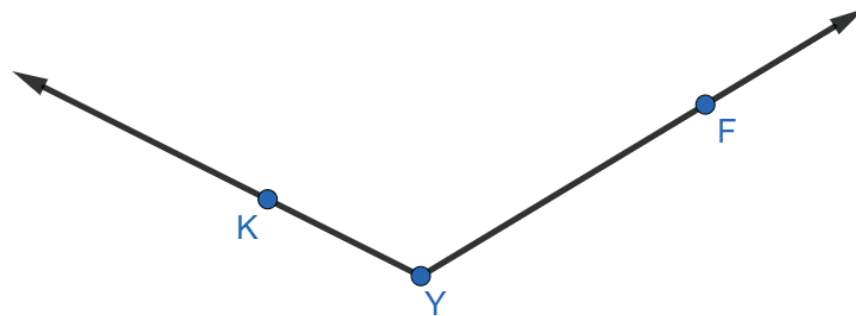


3. $\overrightarrow{BT}$ bisects $\angle HBN$ such that $m\angle HBN = (10x - 2)^\circ$, $m\angle HBT = (5x - 1)^\circ$, and $m\angle TBN = (7x - 23)^\circ$. Find $m\angle TBN$.



1.12.4 Wrap-Up: Ticket Out the Door

1. Construct the angle bisector of $\angle KYF$ using a compass and a straightedge.



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